

Quantifying Ecosystem Services at Jægersborg Hegn



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Abstract

This study investigates how three different management alternatives for the northern part of Jægersborg Hegn affect the provision of ecosystem services, including timber production (expectation value), climate regulation (different carbon estimations), water quantity (average groundwater production), biodiversity (UNA) and recreation (willingness-to-pay). Using the structured-decision-making framework, the three objective based alternatives were evaluated. Alternative 1 had the primary objective of recreation, alternative 2 was a timber production strategy grounded in close-to-nature forest management principles and alternative 3 was biodiversity-centred based on minimal intervention.

All held different strengths and trade-offs. Alternative 1 yielded the highest expectation value and unexpectedly scored highest in the willingness-to-pay. However, it showed reduced UNA score and long-term carbon storage. In contrast, alternative 3 achieved the highest biodiversity and carbon sequestration outcomes, but has no economic value and offered lower recreational value. Alternative 1 performed moderately across all ecosystem services, highlighting the challenge and potential of balancing multiple objectives. Ultimately, this project demonstrates that no single alternative was superior across all ecosystem services. Instead, the most suitable plan depends on management priorities. The consequence table provided an easy and transparent overview of the trade-offs, and is a valuable tool for decision-making in forest planning.

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1. Introduction

1.1. Background

The purpose of this report is to create long-term management alternatives for Jægersborg Hegn. Three selected alternatives will be assessed according to their contribution to ecosystem services and later compared, where key trade-offs will be discussed.

Quantifying ecosystem services can be a helpful tool, because it makes the benefits provided by nature visible, measurable, and comparable. This can help policymakers, landowners, etc. make informed decisions. Furthermore, it allows for weighing different land-use options based on synergies and trade-offs. Ecosystem services often go unaccounted for in traditional forestry systems, so assigning them a value can help justify conservation or restoration (M. King & J. Mazzotta, 2010).

1.2. Need for Research

Denmark was once largely forested, but deforestation for agriculture reduced forest cover from 70% to about 4% over eight centuries. This, combined with other environmental pressures, has accelerated local climate change impacts. Since the late 1800s, rising temperatures and precipitation have affected ecosystems, vegetation, and soils, challenging native species' adaptability and altering forest composition (International Energy Agency, 2023; Wang, 2024).

In response, changing climate, globalization, and societal shifts have broadened the role of forests. Timber production is now complemented by goals such as biodiversity conservation, recreation, natural resilience, and hydrological restoration (Cantarello, 2024).

1.3. Need for Policies

A new forest management strategy should be incentivized with the backing of legal frameworks. The Danish Forestry Act of 2019 and the National Forest Program of 2018 establish this by enforcing mandatory guidelines for emphasis on landscape, environmental protection and socio-cultural history along with the economic investments in a forest (Rigsrevisionen, 2016). The results of these laws and the management plan are evident with a forest cover of 15% in Denmark achieved in 2022 (Miljø- og Ligestillingsministeriet, u.d.). In

addition, the Natura 2000 are EU-directed protected areas that aim to conserve biodiversity by safeguarding and restoring rare, endangered and keystone habitats and ecosystems. As a byproduct, the directive also facilitates sustainable land use practices and enhances ecological resilience and communication among its member countries (Natura 2000, 2024).

1.4. Objectives and Research Question

In Structured Decision Making (SDM), objectives define what matters most- they are the outcomes we care about and the basis for evaluating and comparing different actions. Clearly stating objectives helps ensure transparency, consistency, and accountability in environmental management, especially when navigating complex ecological, social, and economic trade-offs.

Our primary objective is to enhance the recreational value of the northern part of Jægersborg Hegn by creating a nature-based learning environment that fosters deeper public appreciation for the forest's ecological, cultural, and economic dimensions. The forest is envisioned as both a classroom and a space for personal well-being.

Another objective is to increase biodiversity while simultaneously enriching the recreational value of underutilized areas. By supporting ecological diversity and multifunctional land use, we aim to create a forest that serves both conservation goals and community interests. In doing so, we also see potential for low-impact economic benefits tied to local identity, sustainable practices, and partnerships with nearby businesses and producers.

Protecting and enhancing biodiversity remains central to the project. We aim to support forest resilience by maintaining a variety of habitats and highlighting the forest's role in climate regulation, and long-term ecological health.

Although sustainable timber production continues to be part of Jægersborg Hegn's management in two alternatives, this project prioritizes recreational and educational outcomes over the next decade. Ultimately, our goal is to establish Jægersborg Hegn as a model for

integrated forest management- one that balances both ecological integrity and inclusive public engagement.

Research Question: *How can three different management alternatives for Jægersborg Hegn affect the provision of ecosystem services?*

2. Structured Environmental Decision-Making

2.1. Background

With reference to *Structured Decision Making: A Practical Guide to Environmental Management Choices* (R. Gregory et al., 2012), we illustrate the importance of a clear and transparent decision-making framework when addressing complex environmental and natural resource management challenges. These challenges are rarely straightforward—they involve interlinked ecological, social, and economic dimensions that require decisions to be both consistent and accountable.

To evaluate the selected alternatives based on the objectives, we estimated their ecological, social, and economic consequences - explicitly weighing costs and benefits. This allowed us to explore how different actions might affect our priorities, and to make trade-offs visible and manageable. Ultimately, trade-off analysis would help identify the strategy that best advances our stated objectives.

Implementation would then be followed by monitoring, allowing us to learn from real-world outcomes. Through this adaptive management cycle, we remain responsive to change and capable of revisiting objectives and alternatives as new information emerges.

2.2. Key Steps in Structured Decision-Making

We began by clarifying the decision context—defining the project’s purpose, key challenges, and underlying values. The main challenge was to create a multi-functional management plan balancing biodiversity, recreation, and economic goals, which often conflict. We acknowledged the tension between short-term use and long-term conservation.

In the next step, we defined primary and secondary objectives, then selected measurable criteria such as the UNA index, willingness-to-pay, and cost-benefit estimates. To develop alternatives, we chose forest development types (FDTs) based on soil maps and current composition, varying species mix, intensity, and spatial layout—for example, introducing more oak to boost resilience.

We then estimated the consequences of each alternative in terms of ecological resilience, recreational value, and economic return. Using existing species helped reduce disruption and costs.

Finally, we assessed trade-offs—such as enhancing access while protecting habitats, or balancing biodiversity with timber yield. Solutions like orchard meadows emerged as promising compromises, offering ecological, recreational, and economic benefits.

For the sixth step – making a decision and implementing – we discussed possible trade-offs.

3. Presentation of Case Area

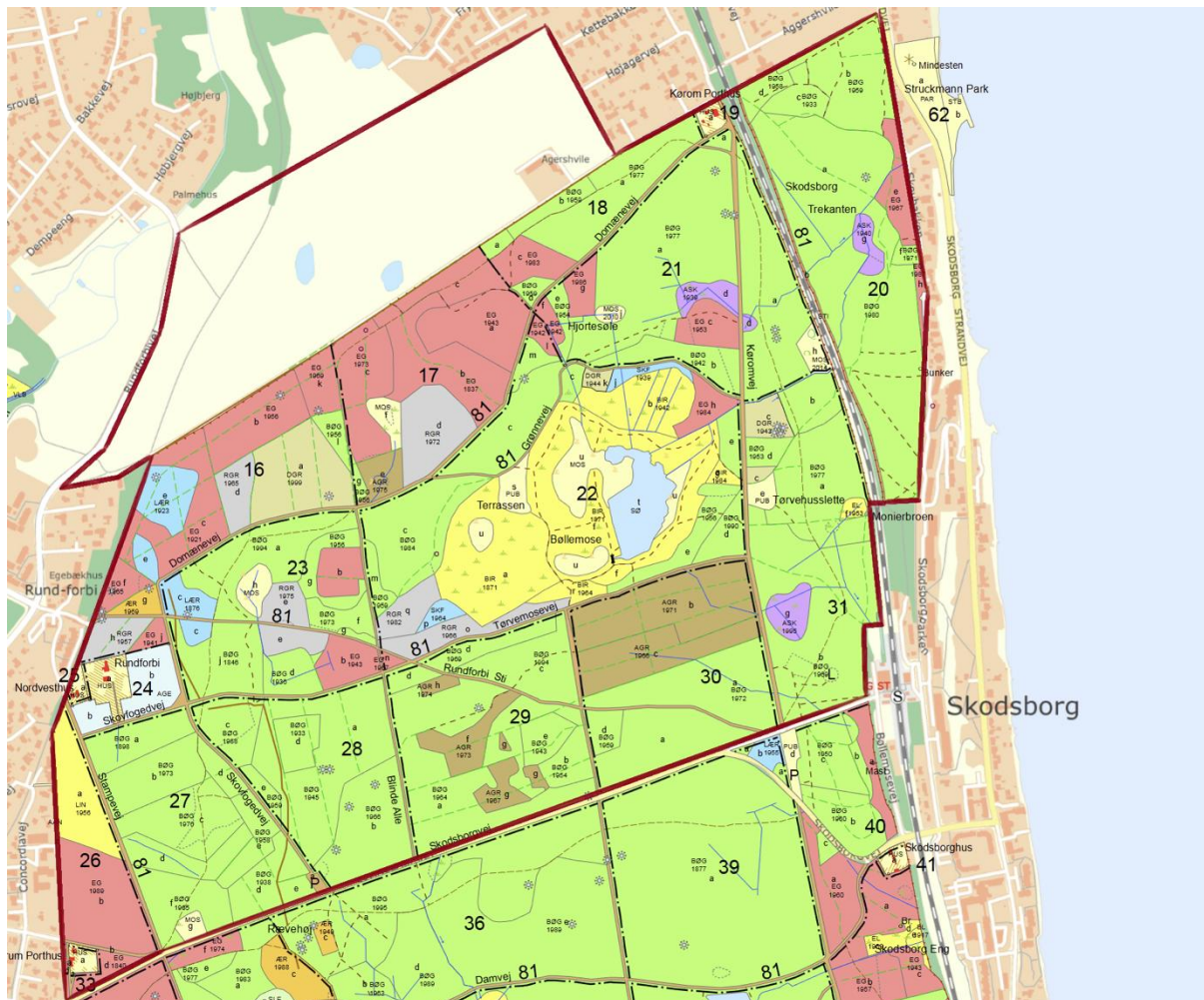


Figure 1: Map of the current management plan and tree species distribution for our study area at Jægersborg Hegn marked by a red line

3.1. History of the Estate

Jægersborg Hegn is a state-owned forest located in Rudersdal Municipality, approximately 10 kilometers north of Copenhagen, Denmark (Naturstyrelsen, n.d.-d). Historically, the estate was owned by the Danish monarchy and was used for parforcehunting (Naturstyrelsen, n.d.-b). In this report, we focus on the northern section of Jægersborg Hegn, marked by the red line in Figure 1, encompassing 188.5 hectares.



Figure 2: Private pictures of some of the recreational activities spotted during our excursion

3.2. Cultural History

Historically, the forest has seen extensive human influence, including 75 legally protected burial mounds. From the 1650s, the area served as royal hunting grounds, and in 1832 it was formally designated as a production forest following its separation from Dyrehaven by a fence. Since then, forestry activities have shaped the structure and species composition of Jægersborg Hegn (Naturstyrelsen n.d.).

3.3. Geological History

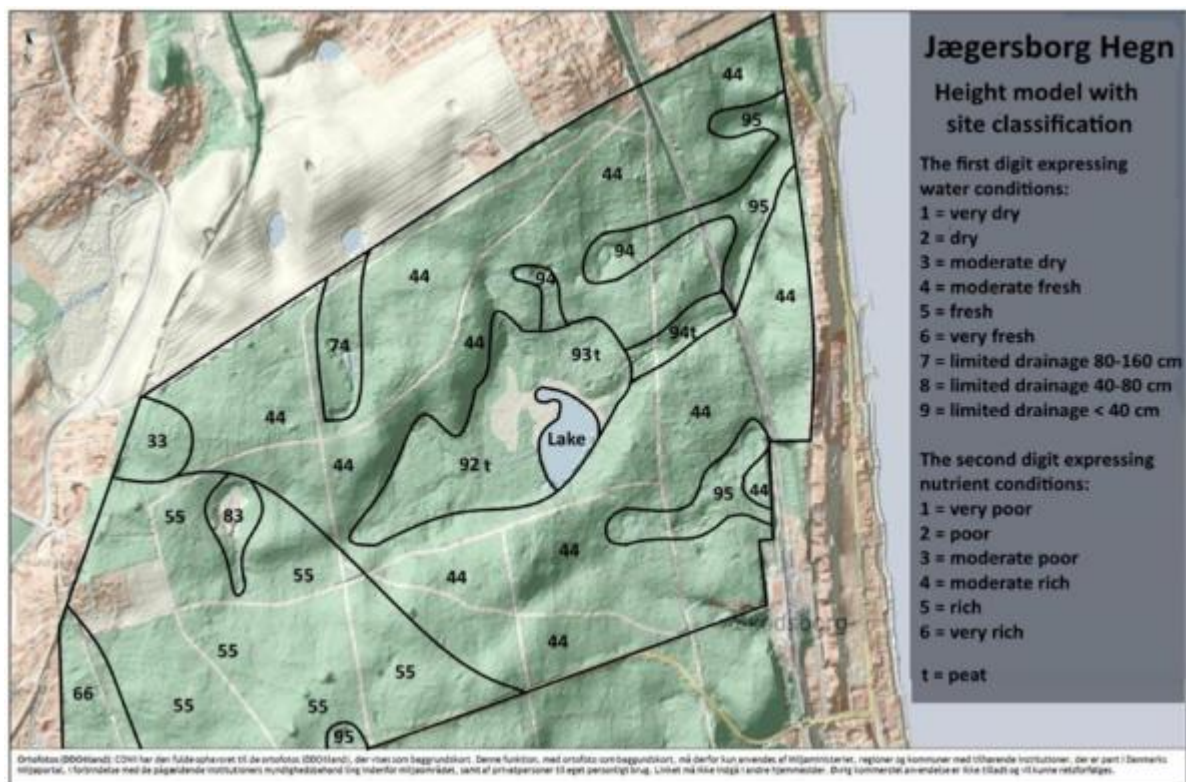


Figure 3: Soil classification map of Jægersborg Hegn

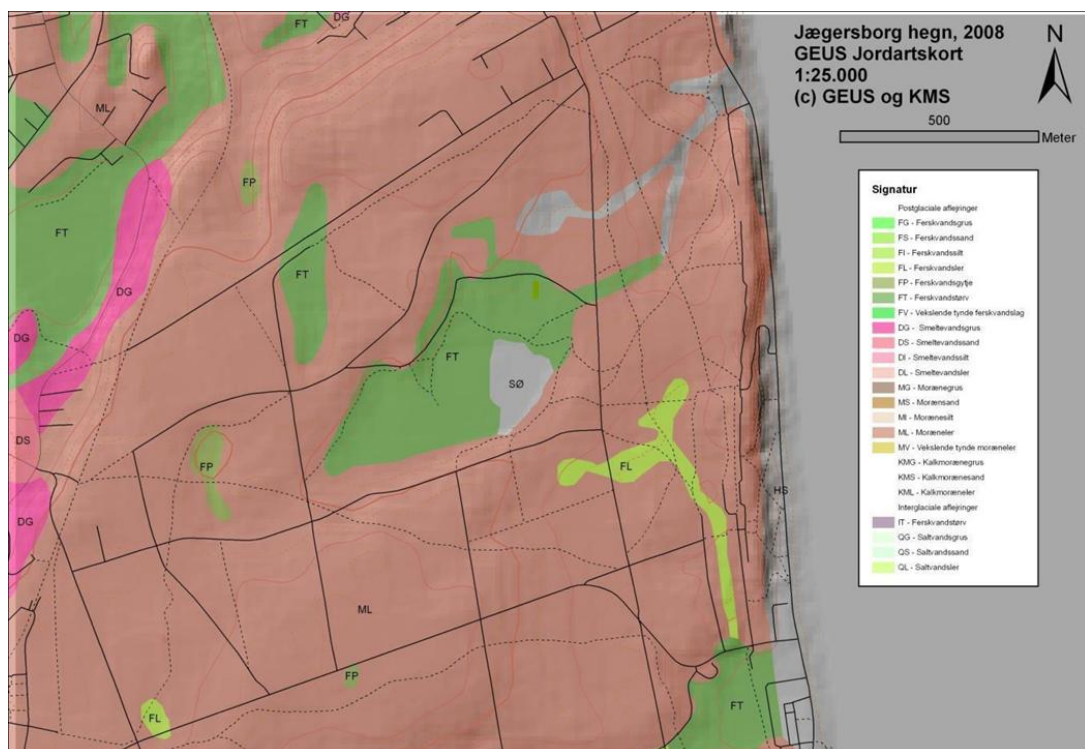


Figure 4: soil type map of Jægersborg Hegn

The soil composition is mainly glacial moraine clay, interspersed with patches of freshwater peat, clay, and gytje, particularly around wetland areas (Figure 3). According to soil classification maps, the area features moderately rich to rich nutrient conditions (soil fertility scores of 4–5), with similar drainage characteristics. In wetter parts of the forest, such as Bøllemosen and surrounding fens, the soil is more waterlogged with limited drainage and nutrient availability (Figure 3).

3.4. Climatic Conditions

The climate in Jægersborg Hegn is classified as Dfb according to the Köppen climate classification, meaning it has a moist continental climate with mild summers and cold winters. The annual average precipitation is 966 mm, and the average temperature is 9.5°C (National Center for Klimaforskning [NCKF], 2024). The average wind direction for Jægersborg comes from the North (Windfinder, n.d.).

3.5. Natural History

According to the stand list, the forest is mainly composed of deciduous species, with *Fagus* dominating the central stands—some dating back to 1835. *Quercus* is prevalent along the northern and western edges, and a few conifer patches remain, including a 25-year-old *Pseudotsuga menziesii* stand. Jægersborg Hegn is also known for its 130-year-old, 35-meter-tall *Larix* trees, selected for seed breeding and with millions of descendants across Europe (Naturstyrelsen, n.d.-a). Many even-aged conifer stands have recently been clear-cut. In some areas, veteran trees were felled to promote biodiversity through microhabitat creation. The forest remains dense due to a low grazer population compared to nearby Dyrehaven.

3.6. Administrative Limitations

Jægersborg Hegn is additionally home to Bøllemosen, a raised bog formed through peat-cutting and drainage, now protected under both Natura 2000 and the Danish Nature Protection Law

§3. Because the area is designated as a Natura 2000 site, the Danish state is required to protect habitats and species listed under the EU Habitats Directive.

3.7. Time-Horizon of the Plan

We have included both a shorter-term and long-term timeline in this report. The short-term time horizon has a 10-year timeframe, which is the so-called tactical plan. Then we also work with a longer time frame of 100 years, which helps us compare our finished conversion to the current state of the Northern part of Jægersborg Hegn.

3.8. Defining Objectives and Performance Measures

The first objective aims to maximize the recreational value of outdoor recreational experiences for the public. Willingness-to-pay (WTP) was selected as a performance measure for recreational value. WTP quantifies the economic value that individuals assign to recreational experiences (Filyushkina et al., 2017, p. 26).

The second objective concerns the enhancement of economic returns, through timber production. This objective represents a tangible and commonly considered criterion in forest and land-use planning. Expectation value (EV) served as the performance measure for economic value, as it reflects anticipated monetary return for each alternative.

The third objective aims to support ecological integrity by conserving species richness and habitat diversity. UNA scores were used as a performance measure for biodiversity. They serve as a proxy for habitat availability, ecological connectivity, and overall biodiversity support within the landscape.

All performance measures provide a standardized metric that makes it possible to compare the different alternatives and identify possible trade-offs between the three.

<i>Fundamental Objective</i>	<i>Means Objective</i>	<i>Performance Measure</i>
Maximize Recreational Value	Enhance forest aesthetics and openness Promote structural and seasonal diversity	Willingness to Pay (WTP)

Maximize Economic Value	Optimize species composition for timber yield Improve timber quality via regular thinning	Expectation Value (EV)
Maximize Biodiversity	Allow natural succession/minimize intervention Retain deadwood and structural complexity	UNA Score

Table 1: Table of fundamental objectives

Other relevant objectives in this report include maximizing groundwater production and maximize carbon sequestration. The performance measures used for these include average groundwater production (m³/ha/year) and carbon storage (Mg/ha/year).

3.9. Description of Alternatives

In this section, we present three strategic forest management alternatives that represent different approaches to meeting the defined objectives of the plan: maximizing recreational value, maximizing economic value and maximizing biodiversity. Each alternative is based on distinct Forest Development Types (FDTs) and associated silvicultural strategies. In all three alternatives, the bog areas within the Natura 2000 network are left unmanaged from a forestry perspective. These zones are strictly protected and excluded from timber harvesting and silvicultural interventions. However, existing low-impact recreational infrastructure, such as walking paths and signage, remains in use to support nature experience and education. No further development is planned, and activities in these areas will be managed to ensure minimal ecological disturbance and compliance with Natura 2000 conservation objectives.

<i>Alternative</i>	<i>Main Objective</i>	<i>FDT Applied</i>	<i>Silvicultural Actions</i>	<i>Expected Outcomes</i>
Recreation	Maximize visitor value	FDT 11 (Beech)	Uneven-aged structure with group selection; light	High visual appeal, scenic walking paths,

			thinning for visual openness; natural regeneration of beech	and seasonal variation (e.g. anemone bloom)
Recreation	Maximize visitor value	FDT 22 (Oak, lime, beech)	Selective thinning to maintain mixture and structure; encourage natural regeneration of oak and lime	Dense, mixed canopy with year-round visual interest; moderate biodiversity
Recreation	Maximize visitor value	FDT 94 (Unmanaged)	No intervention; retention of deadwood and natural disturbances to shape structure	High structural and habitat diversity, including deadwood and uneven-aged stands
Recreation	Maximize visitor value	Orchard Meadow	Establishment of fruit and habitat trees on grassland; grazing; mowing for meadow maintenance	Open, culturally rich landscape with opportunities for passive recreation and biodiversity
Biodiversity	Leave nature untouched	FDT 94 (Unmanaged)	No intervention; allow natural succession and gap dynamics; monitor structural diversity	Reference area for natural processes; high ecological value with low disturbance
Timber	Economic yield	FDT 14 (Beech + spruce)	Mixed planting (e.g. Douglas fir + beech); regular thinning; pruning for timber quality; 90–150yr rotation	High volume growth and economic return; fast-growing conifers with some native species retention

Timber	Economic yield	FDT 21 (Oak + ash, hornbeam)	Group selection or shelterwood; enrichment planting of oak; long rotation (~120 yrs); weed control in early years	High-value hardwoods; improved site adaptation; long-term timber yield
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Table 2: Strategy Table

3.9.1. Alternative 1: Recreation

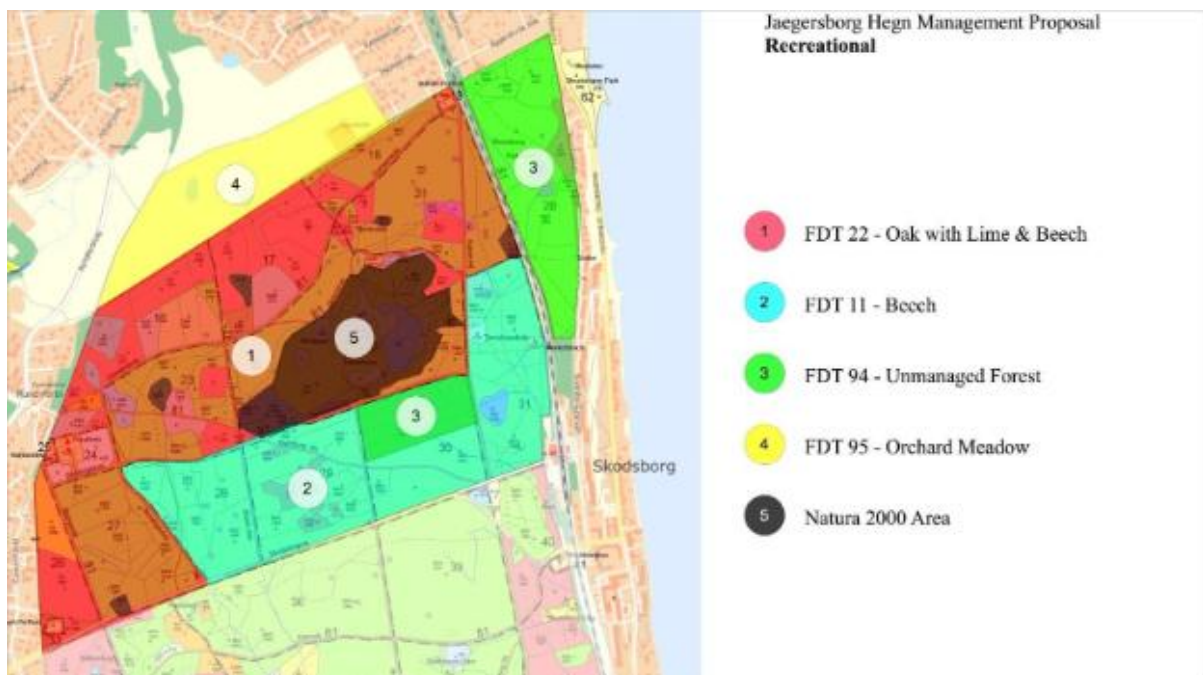


Figure 5: Management proposal map of northern Jægersborg Hegn focusing on recreation

This management alternative enhances recreation while safeguarding ecological and cultural values.

Alternative 1 creates a culturally rich, visually diverse, and accessible landscape by combining four forest development types:

- **FDT 11 (Beech):** Uneven-aged, selectively managed stands with towering trees, a semi-open “pillar hall” feel, and seasonal highlights like wood anemones. Balances recreational and timber objectives (Larsen et al., 2005, p. 9).

- **FDT 22 (Oak with lime and beech):** Denser, species-rich groves offering a sense of seclusion and biodiversity. Balances recreational, ecological, and production goals (Larsen et al., 2005, p. 19).
- **FDT 94 (Unmanaged forest):** Left to natural processes, featuring structural complexity, deadwood, and old-growth areas—ideal for quiet recreation and supporting biodiversity (Larsen et al., 2005, p. 45).
- **FDT 95 (Orchard meadow):** On former farmland, with fruit trees and species-rich grassland. Though unofficial, it enhances landscape character and supports informal recreation like picnicking and walking.

This mosaic of open and closed structures with accessible trails aligns with public preferences—walking is the most popular forest activity in Denmark, and visitors appreciate semi-natural appearances, structural diversity, and seasonal interest (Jensen, 1999; Jensen & Koch, 2004; Skov-og Naturstyrelsen, 2002). By reflecting these preferences, the alternative is expected to boost visitation and strengthen people’s connection with the forest.

3.9.2 Alternative 2: Timber Production within the framework of CTNFM

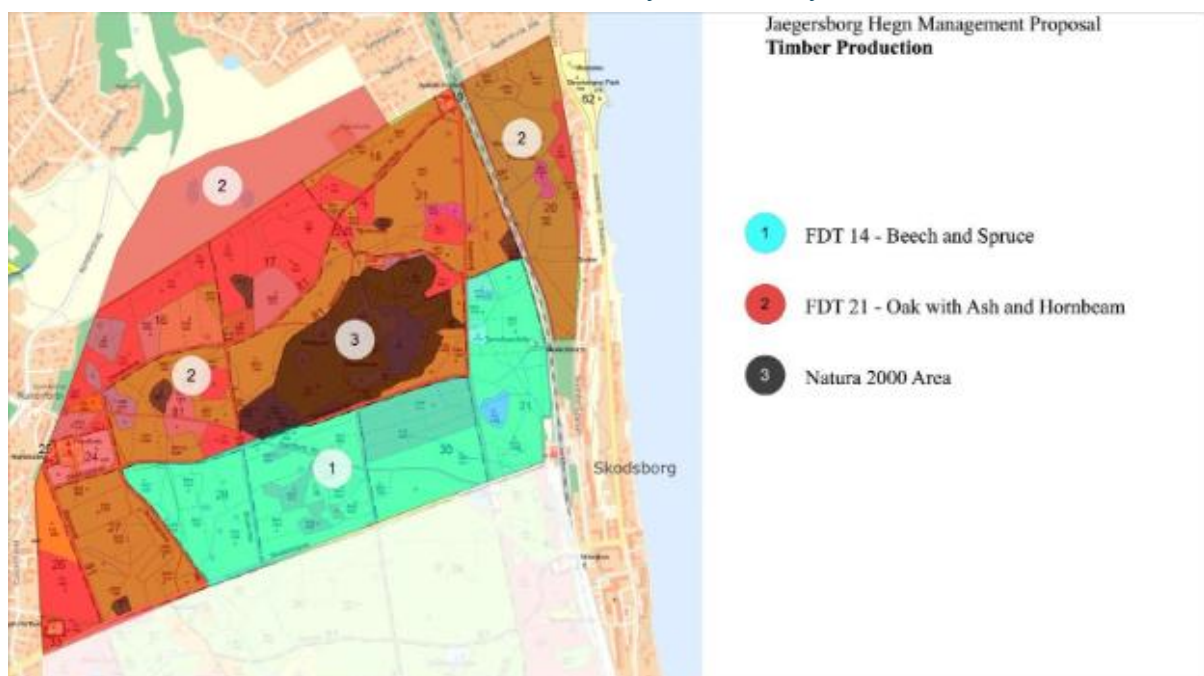


Figure 6: Management proposal map of northern Jægersborg Hegn, focusing on timber production

This alternative emphasizes high-quality timber production while maintaining structural complexity, continuous canopy cover, and long-term forest health. It follows the principles of

Close-to-Nature Forest Management (CNFM), which promote ecological stability, natural regeneration, and the avoidance of clear-cutting, aiming to balance economic and ecological values.

Two forest development types (FDTs) are applied:

- **FDT 14 (Beech with conifers)** is implemented on productive sites. Mixed stands of native beech and conifers like Douglas fir or Norway spruce are managed through group selection or target-diameter harvesting, maintaining continuous cover and structural diversity. Its primary goal is timber production (Larsen et al., 2005, p. 15).
- **FDT 21 (Oak with ash and hornbeam)** focuses on high-value oak over ~120-150-year rotations. Single-tree selection, enrichment planting, and thinning from below support oak regeneration, while ash and hornbeam enhance structure and provide earlier yields. Natural regeneration is prioritized, supplemented by underplanting where needed. Deadwood and habitat trees are retained to support biodiversity, and species are selected for climate and pest resilience. The main objective is timber production (Larsen et al., 2005, p. 17).

This approach supports long-term timber yield, ecological resilience, and a natural forest appearance.

3.9.3. Alternative 3: Biodiversity

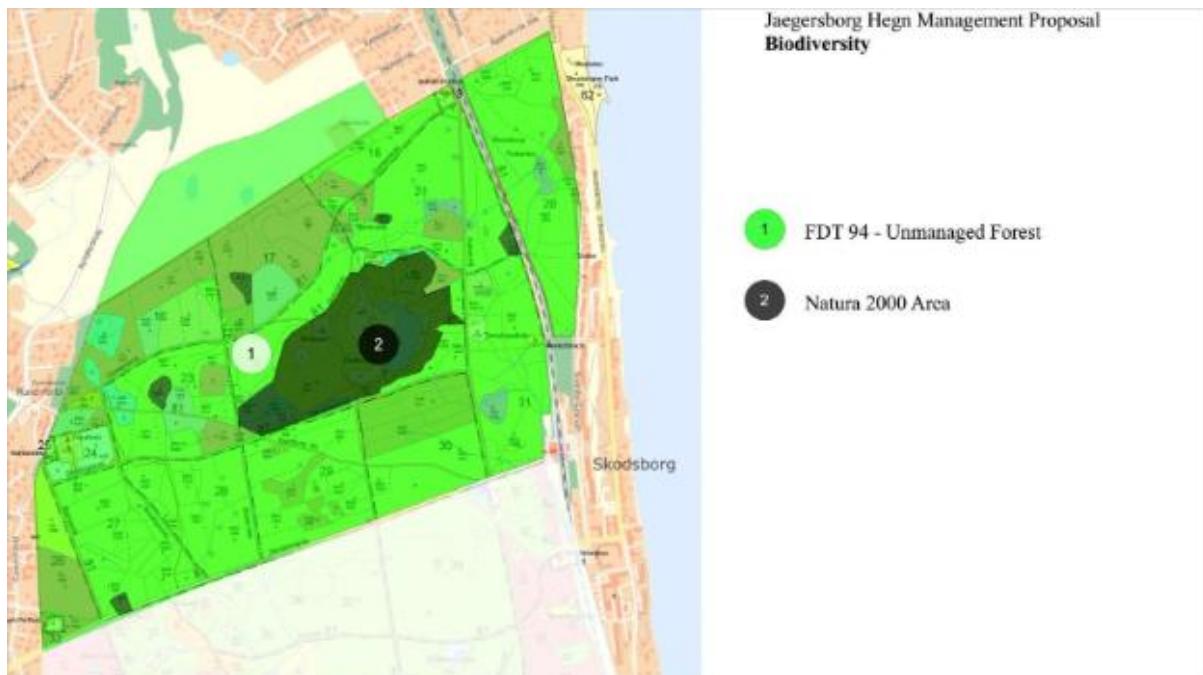


Figure 7: Management proposal map of northern Jægersborg Hegn, focusing on biodiversity

The entire area under this strategy is assigned to FDT 94 (Unmanaged Forest). This forest development type is characterized by natural dynamics, including gap-phase regeneration, deadwood accumulation, and the gradual formation of uneven-aged structures. No planting, thinning, or harvesting is carried out. Instead, ecological processes such as windthrow, senescence, and competition shape the forest over time.

Management is minimal and limited to passive monitoring, except for maintenance of the walking paths. Deadwood, including standing and fallen trees, is left in place to provide habitat for fungi, invertebrates, birds, and small mammals. Veteran trees and microhabitats are preserved. Over decades, these stands are expected to develop old-growth features such as multi-layered canopies, large coarse woody debris, and a mosaic of successional stages. According to the sustainability triangle, FDT 94 has the primary objective of biological value, but also on recreational value (Larsen et al., 2005, p. 45)

4. Timber Production

Timber production is a cornerstone of modern forestry, providing essential raw materials and spurring economic development while advancing sustainable forest management practices

(Smith, 2024). Using advanced silvicultural techniques such as selective logging, timber harvesting not only maximizes yield but also preserves soil integrity and hydrological cycles, thereby maintaining overall forest health (Griffin, 2024).

4.1. Methods

4.1.1 Inventory and Sampling Approaches

Field inventories were updated during two days of forest excursions. In each stand, 3–4 circular plots were randomly selected based on stand size, with plot dimensions chosen to include 15–20 trees. All trees with a DBH over 5 cm were recorded by species, and 3–4 trees per plot were measured for height to capture stand age and species diversity.

After fieldwork, data were tabulated and averaged by species and age class. Stem numbers were extrapolated to stand level, and DBH, height, basal area, and volume were calculated. These values were then added to the forest inventory and compared with previous data and model predictions.

4.1.2 Prediction of Growth, Thinning and Final Harvests

The forest inventory for the northern compartments was filtered out from the common stand list. Further segregation was done according to the compartments as designed in our alternatives. Using this inventory, the species-specific parameters could be obtained for each compartment of each alternative. These included the land-use areas & area-fractions, the age averages, the mean heights, the mean diameters, the mean stem numbers, the mean volumes and the mean basal areas. Weighted compartment-wide means and the land-use areas of conifers versus broadleaves were also deduced alongside these calculations.

This data was then fed into the VIDAR software. Some constraint assumptions were applied while calculating the growth and yield tables. They were as follows:

- The volumes were only computed for trees > 5cm in DBH
- The thinning regimes and yield classes were assigned based on comparison of the software results with the growth and yield tables found in ‘Volume 22 of Contributions from the Northwest German Forest Research Institute’ (Albert. M., 2021) (this database

was used due to the lack of publicly available contemporary Danish growth and yield records)

4.1.3 Forest Development Types

The species composition for the FDTs were formulated in accordance with the guidelines provided in Larsen et. al. (2005). The species composition was decided as follows:

<i>Alternative</i>	<i>FDT</i>	<i>Species</i>	<i>Species composition</i>
Recreation	11	Fagus sylvatica	70%
		Quercus sp.	10%
		Acer pseudoplatanus	5%
		Prunus padus	5%
		Pinus sylvestris	5%
		Picea abies	5%
	22	Quercus sp.	60%
		Fagus sylvatica	20%
		Tilia cordata	10%
		Populus tremula	5%
		Sorbus aucuparia	5%
	95 (Orchard Meadow)	Malus domestica	40%
		Pyrus communis	25%
		Prunus domestica	20%
		Prunus avium	10%
		Juglans regia	5%
Timber Production	14	Fagus sylvatica	40%

		Picea abies	20%
		Quercus sp.	10%
		Acer pseudoplatanus	5%
		Larix decidua	5%
		Abies alba	5%
		Abies grandis	5%
		Pseudotsuga menziesii	5%
		Picea sitchensis	5%
	21	Quercus sp.	50%
		Fraxinus Excelsior	10%
		Carpinus betulus	10%
		Corylus avellana	10%
		Acer platanoides	10%
		Tilia cordata	5%
		Prunus padus	5%

Table 3: Species composition in the compartments

4.1.4 Data Processing

As the models and data availability in the consecutive steps offered lesser flexibility to perform a comprehensive analysis for each of the observed tree species, it was necessary to group the species in terms of their similarities in growth patterns and timber prices. The grouping was done as follows:

Tree Species	Closest matching tree species
<i>Abies alba</i>	<i>Pseudotsuga menziesii</i>
<i>Abies grandis</i>	<i>Pseudotsuga menziesii</i>
<i>Acer platanoides</i>	<i>Fagus sylvatica</i>

<i>Acer pseudoplatanus</i>	<i>Fagus sylvatica</i>
<i>Alnus rubra</i>	<i>Pseudotsuga menziesii</i>
<i>Betula pubescens</i>	<i>Fagus sylvatica</i>
<i>Carpinus betulus</i>	<i>Fagus sylvatica</i>
<i>Corylus avellana</i>	<i>Fagus sylvatica</i>
<i>Fagus sylvatica</i>	<i>Fagus sylvatica</i>
<i>Fraxinus excelsior</i>	<i>Fagus sylvatica</i>
<i>Larix decidua</i>	<i>Pseudotsuga menziesii</i>
<i>Picea abies</i>	<i>Picea abies</i>
<i>Picea sitchensis</i>	<i>Pseudotsuga menziesii</i>
<i>Pinus sylvestris</i>	<i>Pseudotsuga menziesii</i>
<i>Populus tremula</i>	<i>Fagus sylvatica</i>
<i>Prunus padus</i>	<i>Fagus sylvatica</i>
<i>Pseudotsuga menziesii</i>	<i>Pseudotsuga menziesii</i>
<i>Quercus sp.</i>	<i>Quercus sp.</i>
<i>Sorbus aucuparia</i>	<i>Fagus sylvatica</i>
<i>Tilia cordata</i>	<i>Fagus sylvatica</i>

Table 4: Grouping of tree species

The result of the groupings provided the FDT species compositions in terms of feasible and complete input for the conversion models. The revised FDT compositions were as follows:

<i>Alternative</i>	<i>FDT</i>	<i>Species</i>	<i>Species composition</i>
Recreation	11	<i>Fagus sylvatica</i>	80%
		<i>Quercus sp.</i>	10%
		<i>Pseudotsuga menziesii</i>	5%
		<i>Picea abies</i>	5%
	22	<i>Quercus sp.</i>	60%
		<i>Fagus sylvatica</i>	40%
Timber Production	14	<i>Fagus sylvatica</i>	45%
		<i>Pseudotsuga menziesii</i>	25%
		<i>Picea abies</i>	20%
		<i>Quercus sp.</i>	10%

	21	Quercus sp.	50%
		Fagus sylvatica	50%

Table 5: Used species composition

4.1.5 Conversion Models

We opted to use a two-phase conversion model to predict the transition from the existing compartment structure to the new management regime dictated by each of our alternatives. This meant using detailed field data along with cost-effective modelling techniques to obtain accurate extrapolated results. To do this, the growth and yield tables for the converted stands were created by VIDAR using the initial stand parameters as outlined in the forest stand list inventory. Then, all the growth and yield tables were integrated in the allocated spaces for existing and converted compartments respectively in the conversion sheet. The rotation ages for each stand were borrowed from the *Larsen et. al. (2005)* literature and were fed into the spreadsheet alongside the shares of the species composition for each compartment.

4.2. Results

4.2.1. 10-Year Revenue

	<i>Expected discounted revenue in 10 years in each FDT (DKK)</i>	<i>Total expected discounted revenue in 10 years (DKK)</i>
Alternative 1 Recreation	FDT 22 = 0 FDT 11 = 1.092.879 FDT 94 = 0 FDT 95 = unknown	1.092.879
Alternative 2 Timber	FDT 14 = 1.798.940 FDT 21 = 1.499.600	3.298.549
Alternative 3 Biodiversity	FDT 94 = 0	0

Table 6: 10-year Revenues

4.2.2. Revenue at Rotation Age

	<i>EV in each FDT (DKK)</i>	<i>Total EV at time 0 (DKK)</i>
Alternative 1		

Recreation	FDT 22 = 2.766.860 FDT 11 = 2.740.778 FDT 94 = 0 FDT 95 = unknown	5.507.638
Alternative 2 Timber	FDT 14 = 4.632.256 FDT 21 = 3.324.998	7.957.254
Alternative 3 Biodiversity	FDT 94 = 0	0

Table 7: Revenues at rotation ages

4.2.3. Orchard Meadow: Establishment and Management Costs (per ha)

<i>Category</i>	<i>Cost Type</i>	<i>Estimated Cost (DKK/ha)</i>
Site preparation	One-time	3,500
Tree planting (60 trees @ 250 DKK/tree)	One-time	15,000
Tree protection (60 guards @ 100 DKK/tree)	One-time	6,000
Grassland establishment	One-time	2,000
Grazing setup (fencing, water, shelter)	One-time	5,000
Ecological structuring (deadwood, shrubs)	One-time	3,000
Early weed control & mulching (2 years)	One-time	3,000
Total One-time Costs		37,500 DKK/ha
Annual maintenance (mowing, pruning)	Annual	4,000
Annual grazing management (outsourced)	Annual	1,500
Total Annual Costs		5,500 DKK/ha/year

<i>Category</i>	<i>Revenue Type</i>	<i>Estimated Revenue (DKK/ha)</i>
Fruit production (60 trees, low-intensity)	Annual	3,000
Grazing lease income	Annual	1,000
Total Annual Revenue		4,000 DKK/ha/year

Table 8: Orchard Meadow Costs & Revenue

4.3. Discussion

The second alternative, with a focus on timber production, had the highest EV. The FDTs were chosen with an economic profit in mind, meaning that the result is as expected. When choosing the FDTs for the recreational alternative (alternative 1), we also chose FDTs that did well in the wood production part of the sustainability triangle. Despite the broader focus, there still is a notable difference of 3.423.088 DKK in the EV between the two alternatives. The third alternative, which is untouched, is set to 0 DKK as there will be no harvesting of timber.

In this economic model, we chose the same thinning regime for both calculated alternatives. We could also have chosen to do a heavier thinning in alternative 2, where the focus on timber production was even higher.

Furthermore, it is important to note that the EV is based on predictions of the future in terms of timber pricing, cost, subsidies and so on. There are multiple uncertainties when trying to predict the future. Shifts in climate change, policies and natural disturbances can change the ultimate EV (Holopainen et al., 2010).

The orchard meadow is not financially profitable under current conditions due to high establishment costs and low market returns. However, it provides important ecosystem services—such as biodiversity support, pollination, and recreation—which may qualify for subsidies under the EU’s Common Agricultural Policy or national biodiversity programs (European Commission, 2021).

5. Climate Regulation: Carbon Sequestration & Storage

Forests regulate climate by storing CO₂ in biomass, soil, and deadwood. In Denmark, they sequester about 3.5 million tonnes of CO₂ annually—around 7.2 tonnes per hectare—depending on forest type and management (Nord-Larsen et al., 2023). This role is crucial for meeting Denmark’s 2050 climate neutrality goal (Danish Ministry of Climate, Energy and Utilities, 2020).

Carbon pools differ in stability: timber stores carbon for decades, while soil and deadwood are more vulnerable to loss (Poeplau et al., 2017). Both quantity and duration of storage matter.

This section compares carbon storage across three management alternatives—Timber, Recreation, and Biodiversity—over 10 and 100 years.

5.1. Methods

Carbon storage and sequestration were assessed for both the short-term (10 years) and the long-term (100 years) using data provided through course materials and planning spreadsheets.

For the short-term analysis, we used the Excel tool “*Week_21_ForestCarbonManagement_Exercise_Data_Templates_May2025*” introduced in the lectures. Carbon values were calculated based on changes in standing stem volume and harvested volume per hectare. We combined this with area data and conversion values from the Phase spreadsheets to estimate carbon fluxes for each alternative.

The following parameters were used:

- Area size (ha)
- Change in standing stem volume (m³ per 10 years)
- Harvested stem volume (m³ per hectare per 10 years)

For the long-term (100-year) assessment, we used the Phase stand conversion spreadsheets that were also applied in the economic analysis. These provided values for:

- Carbon remaining in the forest
- Carbon extracted
- Carbon stored in deadwood
- Carbon remaining in deadwood over time

The outputs were aggregated and visualized in scatter plots to compare total carbon flows across the three management alternatives: Timber, Recreation, and Biodiversity.

5.2. Results

Short Term:



Figure 9: Illustrated results per hectare

Long Term:

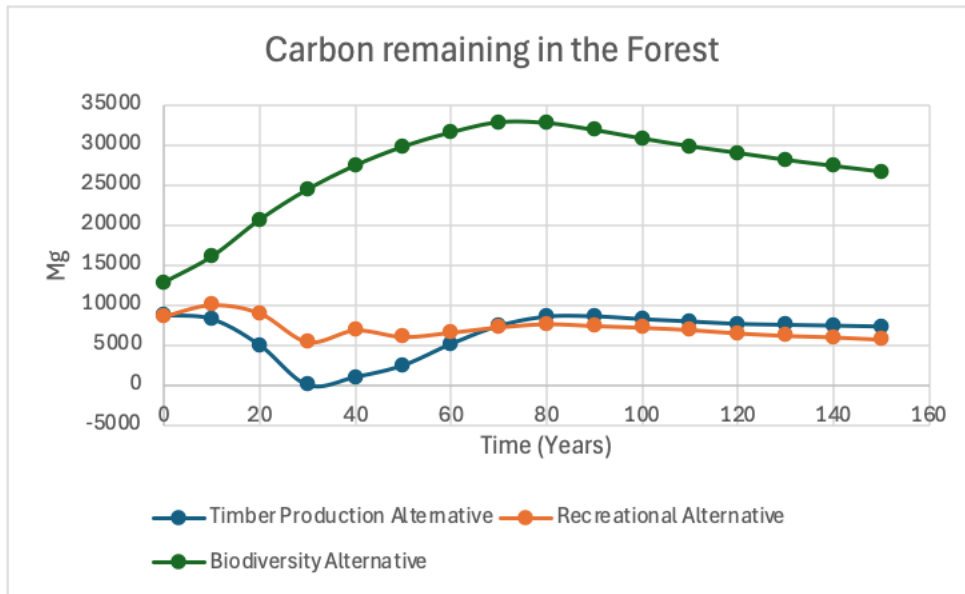


Figure 10: Carbon remaining in the Forest

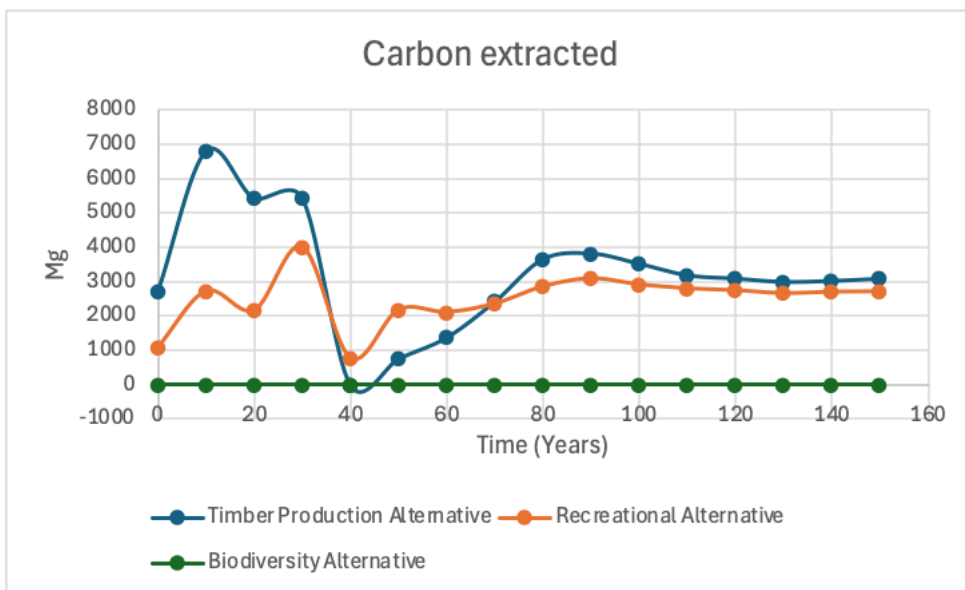


Figure 11: Carbon extracted

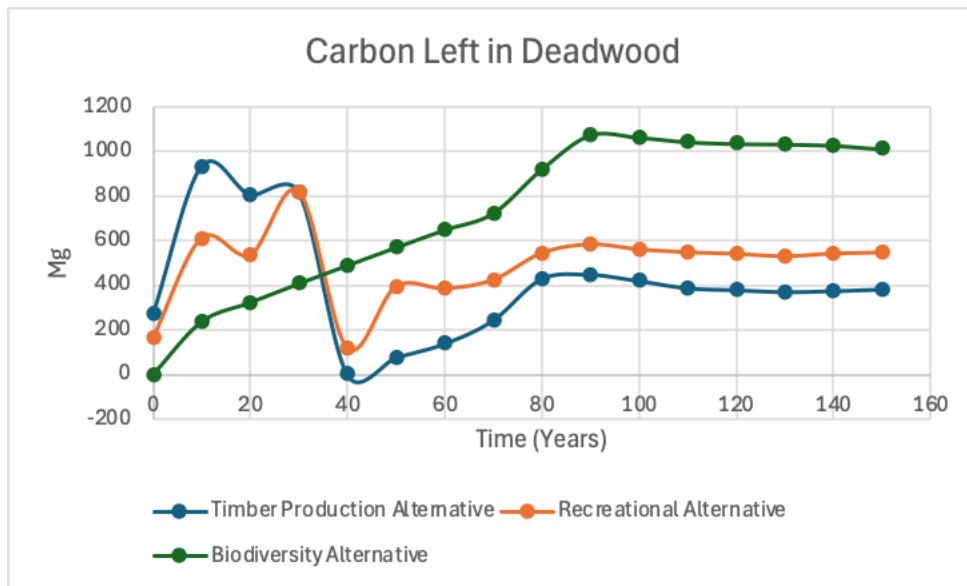


Fig 12: Carbon left in deadwood

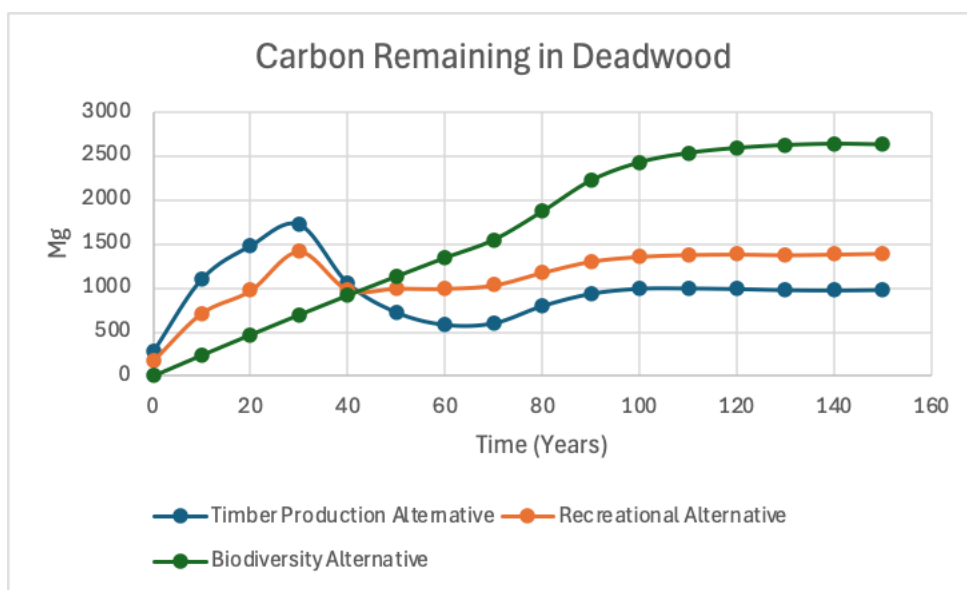


Fig 13: Carbon Remaining in Deadwood

5.3. Discussion

The results reveal distinct carbon outcomes across the three management alternatives. Over 10 years, the Timber alternative extracts the most carbon due to rapid conversion, supporting substitution effects but leaving less carbon stored on site.

The Recreation alternative involves lighter intervention, resulting in lower extraction and higher carbon retention in biomass and deadwood, offering a balanced trade-off between use and storage. The Biodiversity alternative, with no intervention, stores the most carbon—especially in deadwood—and enhances long-term stability, though it provides no substitution benefits. Over 100 years, Timber continues to extract the most but stores the least; Biodiversity retains the most; Recreation remains intermediate. Calculations were based on standardized Excel models, though uncertainties exist around decomposition, growth rates, and harvested wood fate. (Smith et al., 2020; Nabuurs et al., 2018).

In sum, there's a trade-off between carbon use and conservation. A mixed approach with moderate harvesting and retention may best support climate, biodiversity, and recreational goals.

6. Water Quantity

Focusing on water quantity in a management plan is essential because changes in land use significantly alter the soil water balance, which then directly affects groundwater recharge, evapotranspiration, and overall ecosystem health (Salazar et al., 2013, p. 128).

It is key to understand that groundwater recharge is species dependent. A study shows that over a 35-year simulation, spruce forest resulted in significantly lower groundwater recharge, compared to oak forests (Salazar et al., 2013, p. 137). Other factors that influence water dynamics include the climate, stand-age, stand heterogeneity, and management decisions for raising or converting a stand.

6.1. Methods

The seepage values for the individual stands/FDTs were automatically generated in the conversion sheets. These values were relevantly multiplied with the land-use areas of each

species as prescribed in the FDT according to their age-classes. Seepage values of non-forested areas such as fields, clear-cuttings, and recreational areas were also taken into consideration. All the individual seepage values were then summed up to find the total seepage and average groundwater production values.

6.2. Results

	<i>Total Seepage (m3/year)</i>	<i>Average Groundwater Production (m3/ha/year)</i>
Alternative 1 (Recreation focus)	FDT 22 = 77.545,75 FDT 11 = 62.274,10 FDT 94 = Unknown FDT 95 = Unknown Total = 139.819,85	FDT 22 = 1.103,70 FDT 11 = 994 FDT 94 = Unknown FDT 95 = Unknown Average = 1.048,85
Alternative 2 (Timber Production focus)	FDT 14 = 60.155,28 FDT 21 = 99.680,84 Total = 159.836,12	FDT 14 = 861,70 FDT 21 = 1.148,00 Average = 1.004,85
Alternative 3 (Biodiversity focus)	FDT 94 = 161.926,00	FDT 94 = 991,03

Table 9: Groundwater Fluxes

As can be seen in the tables, the groundwater analysis for compartments 3 (FDT 94) & 4 (FDT 95) of Alternative 1 (recreational focus) could not be effectively performed due to the lack of available forest stand-list data and an undefined protocol to analyse the hydrology of FDT 95.

6.3. Discussion

The average groundwater production for the *Quercus* dominated FDTs (21, 22) was higher than those of the *Fagus* dominated FDTs (11, 14) or the unmanaged compartment (FDT 94). A possible reasoning can be formulated that since the water-uptake and storage capacity of the *Quercus* is higher, it leads to a greater hydraulic pressure-gradient in the soil. By the same logic, the higher average groundwater production of the unmanaged compartment of

Alternative 3 can be justified as compared to FDT 11 & 14 compartments (despite them all being *Fagus* dominated), which possibly occurred due to the absence of thinning and therefore a better-established root-system in the compartment.

Despite the partial unavailability of results for Alternative 1, on the whole, the annual groundwater production appears to remain similar for all three alternatives. This is not surprising, considering the heterogeneity of species diversity amongst all the alternatives and the avoidance of clear-cutting in any of the alternatives during their conversion. These results would have potentially differed if the alternatives had promoted monocultures or involved clear-cutting of the forest.

Several uncertainties can be involved during the formulation of hydrology results. These include measurement and computational errors, incomplete datasets, insufficient resolution of spatial and temporal data (including seasonal variations), sampling biases, and underlying assumptions during model building leading to oversimplification of the models.

7. Biodiversity

Denmark's forests are essential for biodiversity, providing habitat for approximately 52% of the country's red-listed species—particularly in structurally diverse areas like old-growth stands, forest glades, and deadwood-rich zones (Nygaard et al., 2021). Although habitat monitoring began under the EU Habitats Directive in 2006, comprehensive data on species-rich, habitat specialist groups remain scarce (Ejrnæs et al., 2008), and the absence of long-term biodiversity data continues to hinder effective conservation (CBD, 2024). To make such initiatives effective, improved monitoring is needed—ideally combining advanced technologies like remote sensing with participatory approaches involving local communities (Biodiversa+, 2022; Danielsen et al., 2010).

7.1. Methods

To assess the current and future biodiversity potential of the northern part of Jægersborg Hegn, we used a pragmatic, proxy-based approach due to time and scale limitations. Instead of full species inventories and detailed habitat mapping, we applied the UNA Forest Structure Index (UNA-FSI)—a scientifically validated tool that estimates biodiversity potential based on

observable structural features like tree species diversity, vertical layering, deadwood, age variation, and ecological continuity (Nygaard et al., 2016). As a class, we conducted inventories in selected compartments and calculated UNA-FSI values based on field data. Since only a subset of compartments could be assessed, we extrapolated the results by assigning average UNA-FSI scores to broader Forest Development Types (FDTs), allowing us to estimate structural biodiversity across the planning area. We then projected changes in UNA-FSI over a 100-year horizon based on expected structural developments from proposed silvicultural strategies. Lastly, we considered biodiversity-promoting practices in temperate forests—like increasing species diversity, preserving old-growth elements, creating light gaps, and encouraging natural regeneration—when designing interventions for each FDT (Lindenmayer et al., 2006; Paillet et al., 2010).

7.2. Results

The results support the idea that biodiversity benefits most under the biodiversity alternative, followed by the recreational alternative, and least under Timber Production. The results are based on UNA averages for the existing stand. The full predicted UNA assessment can be found in the appendix.

7.2.1 Recreational Alternative

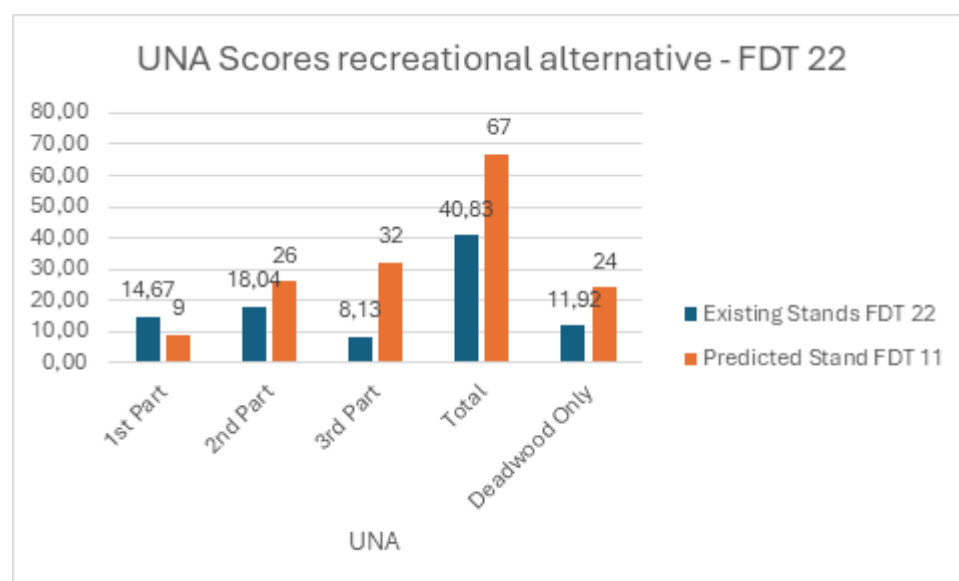


Figure 14: UNA score comparison of existing and future (predicted) stands

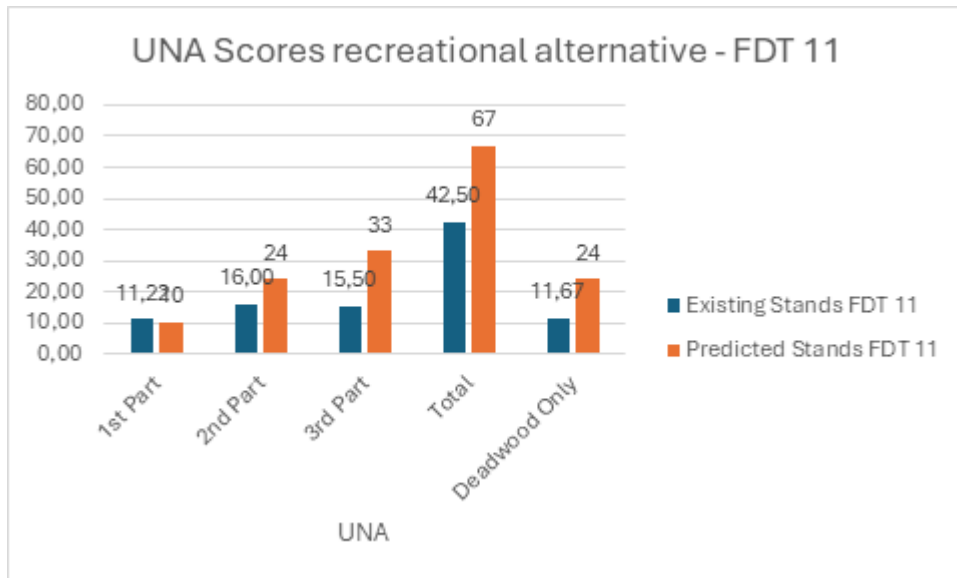


Figure 15: UNA score comparison of existing and future (predicted) stands

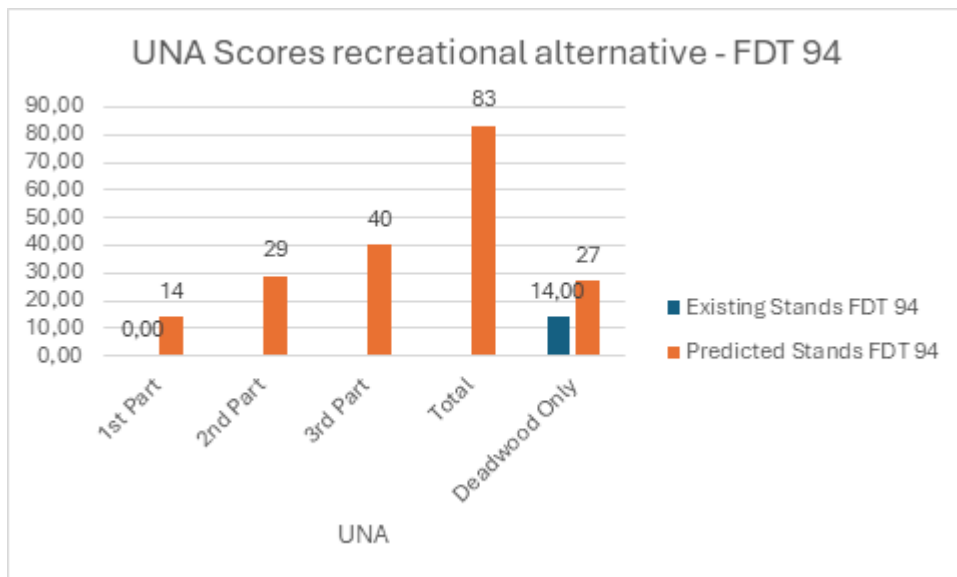


Figure 16: UNA score comparison of existing and future (predicted) stands

7.2.2. Timber Production

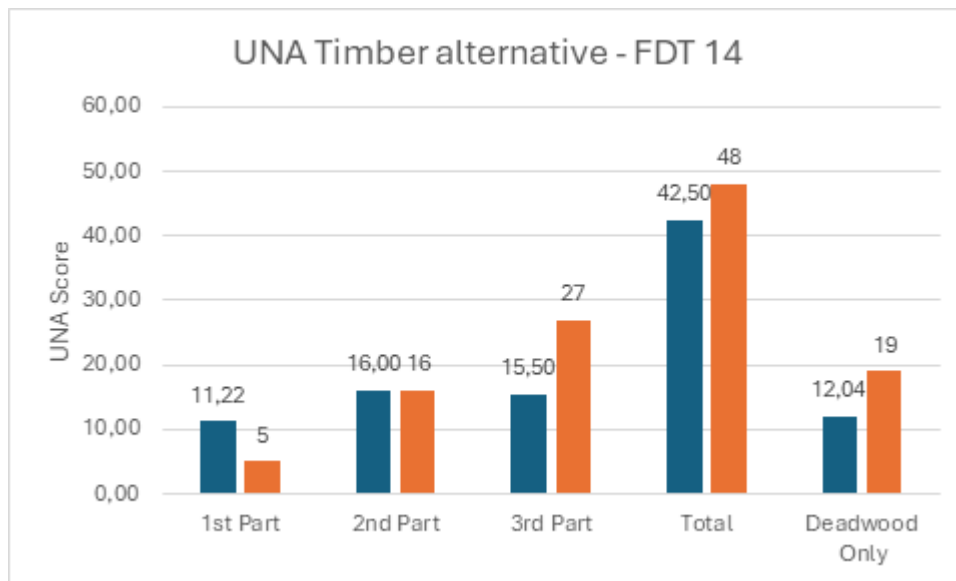


Figure 17: UNA score comparison of existing and future (predicted) stands

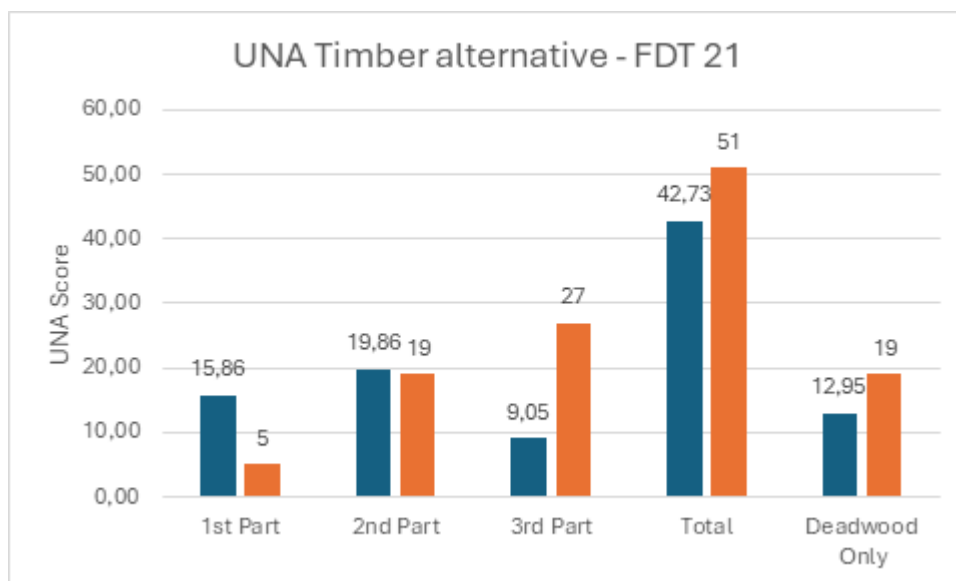


Figure 18: UNA score comparison of existing and future (predicted) stands

7.2.3 Biodiversity Alternative

Tree Species	Group	UNA Index Score Average
Douglas fir, Norway spruce, Larch, Abies grandis	Conifer	27,5

Beech, Oak, Hornbeam, Maple, Lime, Cherry	Deciduous (high quality)	40
Ash, Red alder, Birch, Scots Pine	Deciduous (moderate quality)	No Data
Cypress, Other uses	Misc. conifer	No Data

Table 10: UNA indices

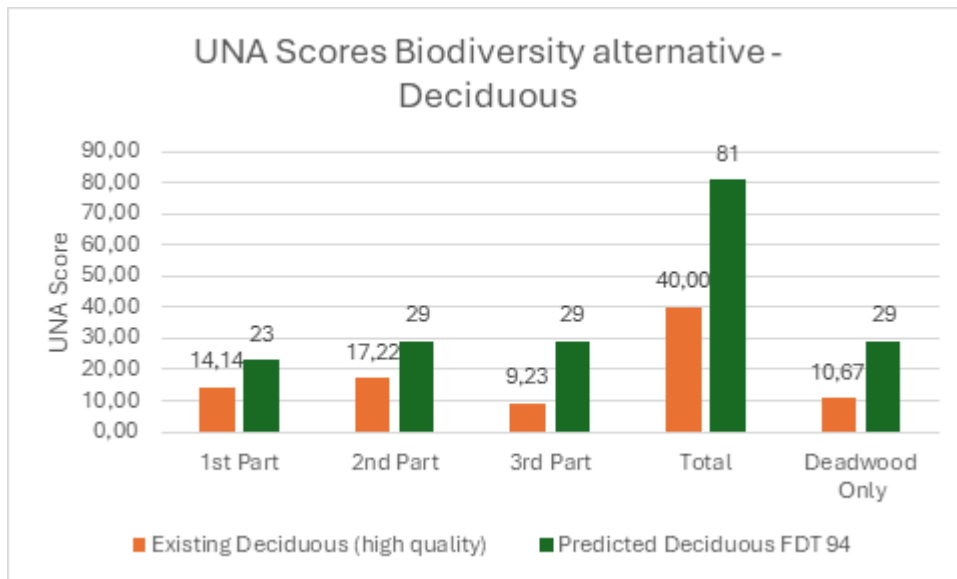


Figure 19: UNA score comparison of existing and future (predicted) stands

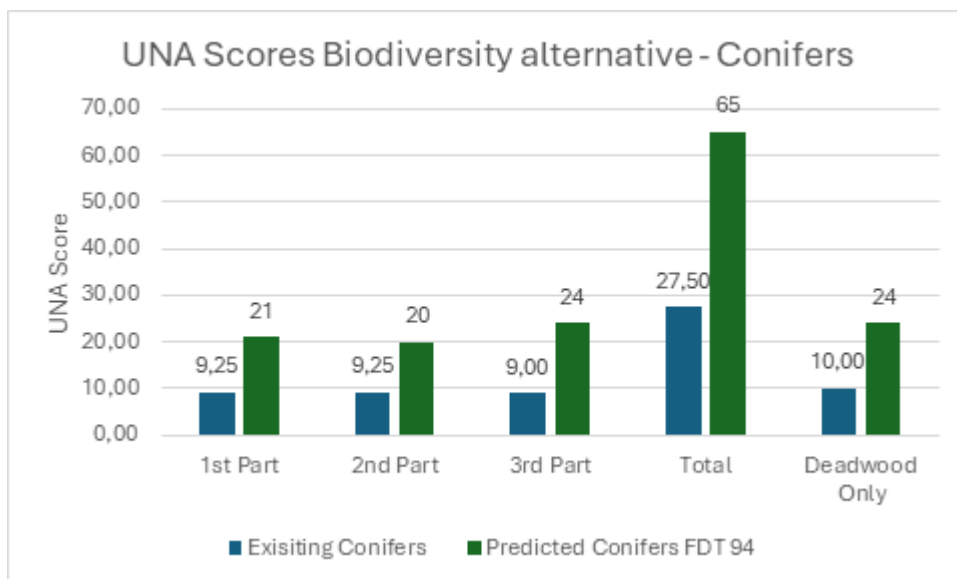


Figure 20: UNA score comparison of existing and future (predicted) stands

7.3. Discussion

The UNA Forest Structure Index reveals clear differences in predicted biodiversity across the three management alternatives.

In the **Recreation alternative**, selective interventions and added infrastructure aim to enhance visual and recreational appeal while retaining ecological structure. Biodiversity remains relatively high, especially in deciduous areas, though harvesting and disturbance may reduce deadwood and regeneration potential (Paillet et al., 2010).

The **Timber alternative**, focused on wood production, involves clear-cutting and thinning, leading to simplified forest structures and lower UNA scores—consistent with studies linking intensive management to reduced biodiversity (Paillet et al., 2010).

The **Biodiversity alternative** assumes no intervention in FDT 94 over 100 years. Here, deciduous stands—especially beech and oak—reached the highest UNA scores (81), reflecting their capacity to develop deadwood, multilayered canopies, and rich microhabitats (Brunet et al., 2010). Coniferous stands also performed well (score: 65), aligning with findings from old unmanaged conifer forests (Christensen et al., 2005).

Still, uncertainties remain, as UNA projections are based on forest types and succession patterns rather than long-term monitoring. Local pressures, climate change, or herbivory could alter outcomes, and the index reflects structural—not species-level—biodiversity (Paillet et al., 2010).

8. Recreation

Recreation is one of the most significant ecosystem services provided by forests. The selected forest management strategy can substantially influence the recreational appeal of a given forest area.

Due to Jægersborg Hegns proximity to the densely populated capital region and the easy accessibility by train, the forest is a popular destination for recreational activities such as walking, biking, and visiting playgrounds and fitness tracks. Together with Dyrehaven, it is

one of the most visited nature area in the country, with more than 7.5 million visitors a year (Naturstyrelsen, n.d.-c).

8.1. Methods

Studies indicate that, within a forest stand, older trees are generally preferred over younger ones. This preference may be attributed to the increased density typically found in younger stands, which can hinder physical access and reduce visual appeal. In contrast, older stands tend to be more open and penetrable (Filyushkina et al., 2017). Moreover, heterogeneous forest structures, characterized by trees of varying heights, are generally favored over homogenous monocultures. In the Danish context, forest visitors show a marked preference for broadleaved or mixed-species forests compared to conifer-dominated ones (Filyushkina et al., 2017).

Based on these assumptions on the preference of Danish forest visitors, we have estimated the willingness-to-pay (WTP) for our different alternatives. Each alternative was evaluated based on the forest development types (FDTs) associated with it. Specifically, each FDTs were assessed on a scale from 0 to 3, according to the dominant tree composition: broadleaved, coniferous, or mixed (approximately 50:50). For each alternative, an average score was calculated across the relevant FDTs. This procedure was repeated to assess tree height.

To evaluate diversity in tree species composition and height across stands, we counted the occurrence of previously assigned scores. Subsequently, we measured the straight-line (bird's-eye) distance from the centers of selected urban areas to the northern part of Jægersborg Hegn, as well as to nearby forest substitutes (Rude Skov, Trørød Hegn, southern Jægersborg Hegn, Ravneholmene, and Folehave). These distances were incorporated into the utility model provided in the course, under the assumption that the average number of visits per inhabitant is 15.1.

8.2. Results

To sum up our results, we calculated the total willingness to pay (WTP) for all of the cities (Nærum, Trørød, Vedbæk, Søllerød, Holte, and Virum) in each alternative. As illustrated by the table below, the calculations show notable differences in recreational value across the three management alternatives. Alternative 2 received the highest scores in WTP, indicating a surprisingly high recreational appeal, despite not being designed within the recreational

objective. For alternative 2, the WTP turned out to be 3.359.455 DKK/visit. Although designed with the primary objective of enhancing recreational value, alternative 1 underperformed and scored second lowest on WTP. Lastly, the third alternative scored the lowest.

<i>Alternative</i>	<i>Total WTP (DKK/visit)</i>
1 - recreation	974.911
2 – timber production	3.359.455
3 - biodiversity	677.403

Table 11: total scores of the recreational model

8.3. Discussion

The high score of alternative 2 can be explained by the higher variation in tree species and height structure between stands, which, according to the study, significantly enhances recreational value. This structured diversity may unintentionally create a more visually interesting and varied forest landscape that appeals to visitors.

Even though the first alternative was created with a goal of maximizing recreational value, it did not obtain the highest score in the recreational model. This apparent contradiction might be explained by the emphasis on just two structural forest attributes: tree species composition and tree height structure between stands. Because our modelling framework captures only two variables, other recreational features purposely included in alternative 1 were invisible to the model. Had the assessment allowed for these other recreational elements, alternative 1 might well have ranked differently. For example, FDT 22 was assessed to have a high recreational value, due to the closeness of the forest. This makes it possible to obtain a feeling of being alone. However, Larsen et. al. also emphasizes that a large portion of the recreational value arises with a variation of forest types as well as open spaces at landscape level (Larsen et al., 2005, p. 9).

The third alternative scored the lowest. While it included structural diversity and ecological richness, it may not offer much visual contrast between the different stands as the timber-focused alternative. Although uneven-aged mixed stands are generally preferred, a lack of variation between stands may reduce the perceived recreational value. If all stands in a forest look similar, the overall recreational experience may appear monotonous. Its biodiversity

benefits also include elements not directly appreciated by the model, such as understory richness and deadwood.

9. Overall Comparison of alternatives and impacts

9.1. Consequence tables of alternative strategies

Fundamental objectives	Performance measure	Alternative 1 (Recreation)	Alternative 2 (Timber)	Alternative 3 (Biodiversity)
Maximize recreational value	Total WTP per. visit across six cities	974.911 DKK/visit	3.359.455DKK /visit	677.403 DKK/visit
Maximize economic value	EV at time 0	4.534.166 DKK	7.957.254 DKK	0 DKK
Maximize biodiversity	UNA	FDT 22 - 67 FDT 11 - 67 FDT 94 - 83	FDT 14 - 48 FDT 21 - 51	FDT 94 (deciduous) - 81 FDT 94 (conifers) - 65

Table 12: Consequence table for our chosen objectives

Other objectives	Performance measure	Alternative 1	Alternative 2	Alternative 3
Maximize groundwater production	Avg. Groundwater production	1.048,85 m3/ha/year	1.005 m3/ha/year	991,03 m3/ha/year
Carbon Sequestration (100 years)	Carbon remaining in the forest	7240,54 Mg	8280,54 Mg	30830,75 Mg
	Carbon extracted	2918,77 Mg/10yr	3530,49 Mg/10yr	0 Mg/10yr
	Carbon left in DW	559,0 Mg	419,01 Mg	1063,24 Mg
	Carbon remaining in deadwood	1359,68 Mg	993,53 Mg	2426,94 Mg

Table 13: Consequence table for other relevant objectives

9.2. Key Trade-offs

The consequence table highlights the distinct priorities and trade-offs of the three alternatives. Alternative 2 (Timber) yields the highest economic value (7.96 million DKK) and recreational value (3,359 DKK/visit), despite its production focus. However, it scores lowest in biodiversity (UNA 48–51) due to structural simplification. Carbon sequestration (8,280.54 Mg) and groundwater production (1,005 m³/ha/year) are moderate. This alternative represents a multifunctional forest model, combining strong economic returns with some ecological and social benefits.

Alternative 3 (**Biodiversity**) performs best in biodiversity (UNA 81 and 65), carbon storage (34,320.93 Mg total), and aligns with conservation targets (e.g., Natura 2000, rewilding). However, it offers no economic return or timber yield, and recreational value is lowest (677 DKK/visit). Groundwater production is also lowest (991 m³/ha/year). This scenario best fits strict conservation areas.

Alternative 1 (**Recreation**) strikes a balance, ranking second in economic (4.53 million DKK), recreational (975 DKK/visit), and biodiversity outcomes (UNA 67–83). It has the highest groundwater recharge (1,049 m³/ha/year) and good carbon performance (7,240.54 Mg), preserving structural features like deadwood and species diversity.

If recreational value is prioritized, Alternative 2 offers the highest benefit. For biodiversity and long-term carbon storage, Alternative 3 is optimal. Alternative 1 provides the most balanced outcome, supporting all objectives moderately well.

9.3. Uncertainties and Implications

The management alternatives and evaluations are based on a lot of assumptions that should be investigated further before implementing the assessed strategies. First, the assumption is that the updated version of the stand list that we have been basing our conversions on is correct.

The second assumption is that the soil/nutrition scale that we have been basing our choices of FDTs on is also correct.

Regarding the economic modelling of the three alternatives, some simplifications were made. In the spreadsheets provided during the course, there was a limited amount of tree species to choose from. Therefore, it was necessary to categorize the other species, primarily based on our own assumptions, which would be needed to investigate further.

The biodiversity assessment was made based on UNA scores. The data collection did not provide a full overview of UNA scores for the entire northern part of Jægersborg Hegn, as we are with in this report. That implies that there might be errors and omissions. Furthermore, our predictions of future UNA scores after conversion are only partly based on scientific research, but still remain assumptions.

WTP was based on tree species and tree height. Other factors, such as easy accessibility or recreational options available in the different nature areas might have provided different results. Other tools used could have been questionnaires and interviews with the visitors, to get a deeper understanding of what they appreciate, when choosing which nature area to visit.

10. Conclusion and reflections

In this project, we explored how three different management alternatives for Jægersborg Hegn affect the provision of ecosystem services, timber production, climate regulation, water quantity, biodiversity and recreation. By applying the SDM framework, we systematically compared a recreation-focused alternative (alternative 1), a timber production strategy rooted in CTFM principles (alternative 2), and a biodiversity-centred approach based on minimal intervention (alternative 3). Each alternative revealed strengths and trade-offs illustrating the multidimensional nature of forest planning.

Alternative 2 provided the highest EV and surprisingly scored highest in the recreational modelling (WTP), due to its structural variety and managed openness. However, this came at the cost of reduced biodiversity (UNA score) and lower long-term carbon storage. In contrast, alternative 3, yielded the greatest gain in UNA scores and carbon storage. Yet, it lacked economic returns and visual variation between stands, reducing its recreational appeal. The recreational alternative (alternative 1) scored second across most ecosystem services, although

it was designed to enhance visitor experience. This illustrates how multifunctionality may require trade-offs but still can deliver broadly acceptable outcomes.

In this project, we learned that tools like EV, UNA and WTP were useful, but sometimes limited. Other benefits, such as educational value, which was a part of our primary objective, was invisible to our WTP model. Moreover, many of our assumptions, such as future timber pricing, species growth and public, carry uncertainties that would need further investigation before real-world implementation.

This analysis demonstrates that no single alternative is superior across all ecosystem services. Instead, the most suitable management plan for Jægersborg Hegn depends on the priorities of decision-makers – whether that be ecological preservation, economic output or public enjoyment. The consequence table provided an easy and transparent basis for navigating these choices serves as a valuable model for forest planning.

This study also revealed important areas for future research. First, a more detailed and long-term biodiversity assessment, such as species richness or abundance, rather than proxies like the UNA index, would strengthen the validity of biodiversity outcomes. Additionally, the valuation of recreational services would be refined through user surveys, interviews or GPS-based visitation data to capture more nuanced visitor preferences, beyond tree structure and species. Economic modelling could be expanded by including potential subsidies for biodiversity and other ecosystem services.

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Appendix

1. Summary of existing stands per management unit

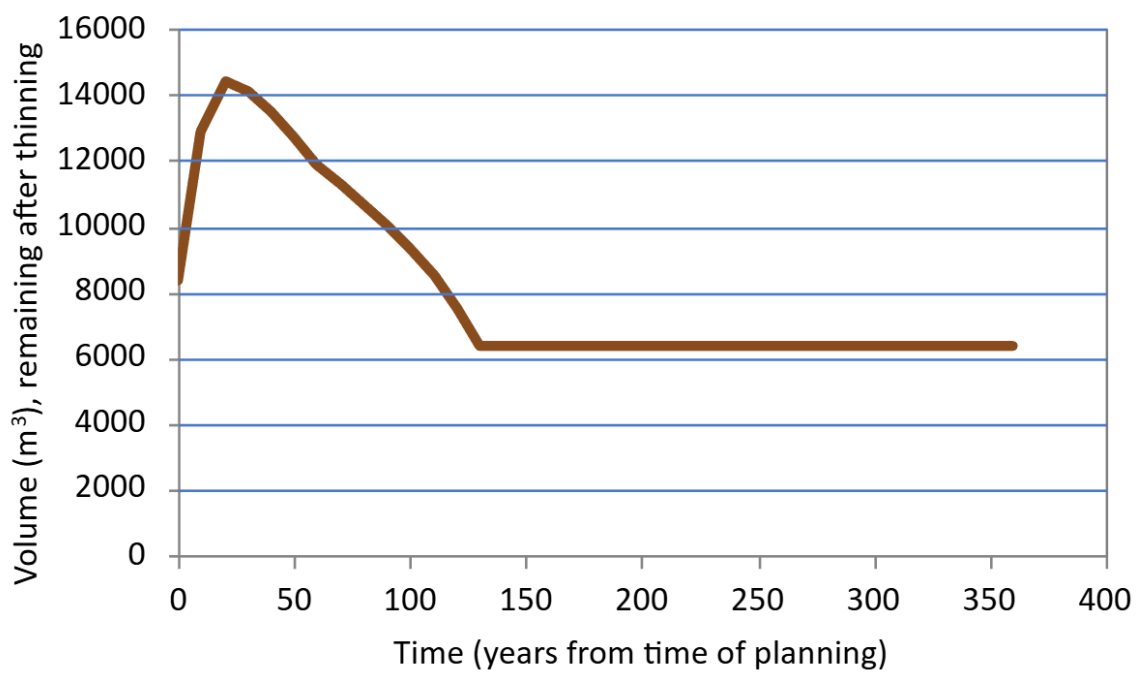
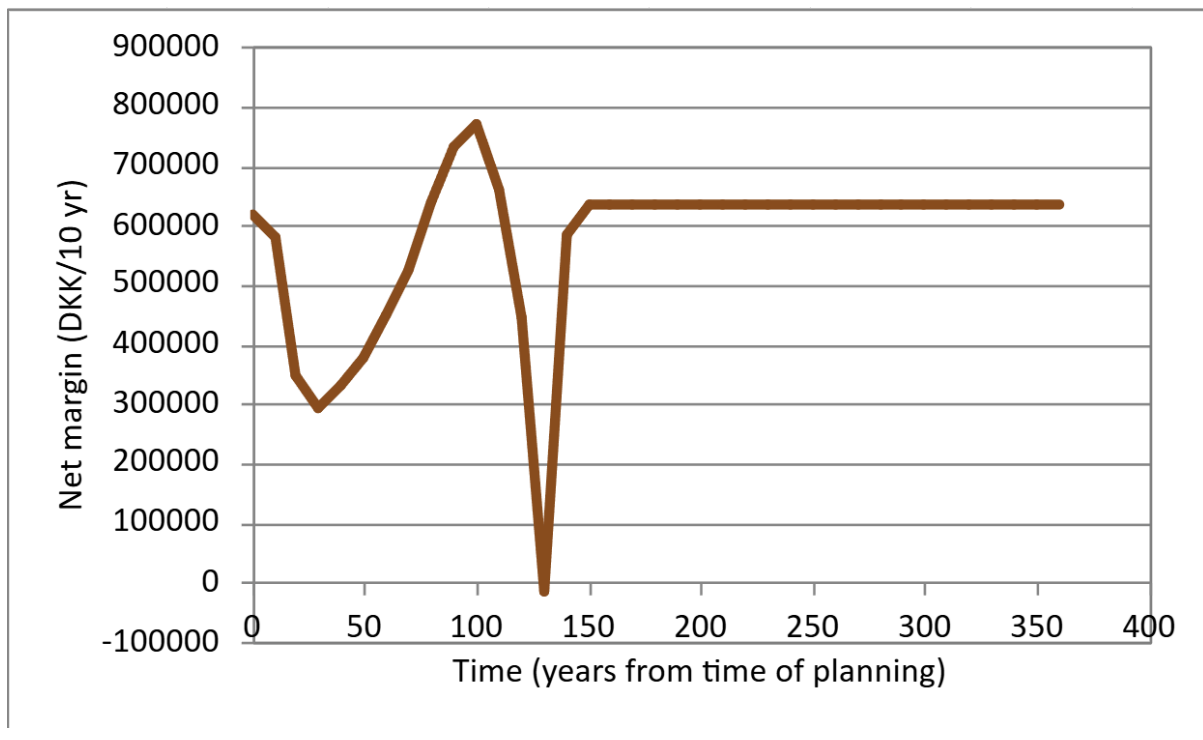
Recreational Alternative				
Unit 1	FDT 22	Land Use	Area (ha)	Age Average Species
		Douglas fir	1,176	32,25
Area Conifers (ha)	5,73	Norway Spruce	3,3765	85,25
		Beech (Old)	9,8995	145,11
Area Broadleaves (ha)	33,11	Beech (young)	17,7275	52,79
Other Uses (ha)	26,55	Hornbeam	0	-
		Oak	13,561	86,55
Total area (ha)	65,39	Maple	1,1865	57,00
		Larch	0,877	121,00
Average tree age (years)	80,70	Scots Pine	0,068	74,33
		Cypress	0,234	-
		Cherry	0,638	14,00
Unit 2	FDT 11	Land Use	Area (ha)	Age Average Species
Area Conifers (ha)	1,84	Douglas Fir	0,79	85,00
Area Broadleaves (ha)	32,81	Norway Spruce	0,389	50,50
		Beech (Old)	8,713	151,25
Other Uses (ha)	28,01	Beech (young)	30,837	52,79
Total Area (ha)	62,65	Abies grandis	0,656	56,67
		Oak	1,133	87,50
Average tree age (years)	87,33	Ash	0,66	59,00
		Red Alder	0,18	73,00
Unit 3	FDT 94	Land use	Area (ha)	Age Average Species
Area Conifers (ha)	6,60	Maple	3,718	54,50
Area Broadleaves (ha)	17,91	Norway Spruce	1,484	59,50
		Beech (Old)	0,982	136,75
Other Uses (ha)	0,98	Beech (young)	12,672	60,00
Total area (ha)	25,49	Abies grandis	5,116	59,50
		Oak	1,23	50,67
Average tree age (years)	75,20	Ash	0,144	86,00
		Red Alder	0,144	72,00

Figure A: Summary of existing tree species and their age for the future management units

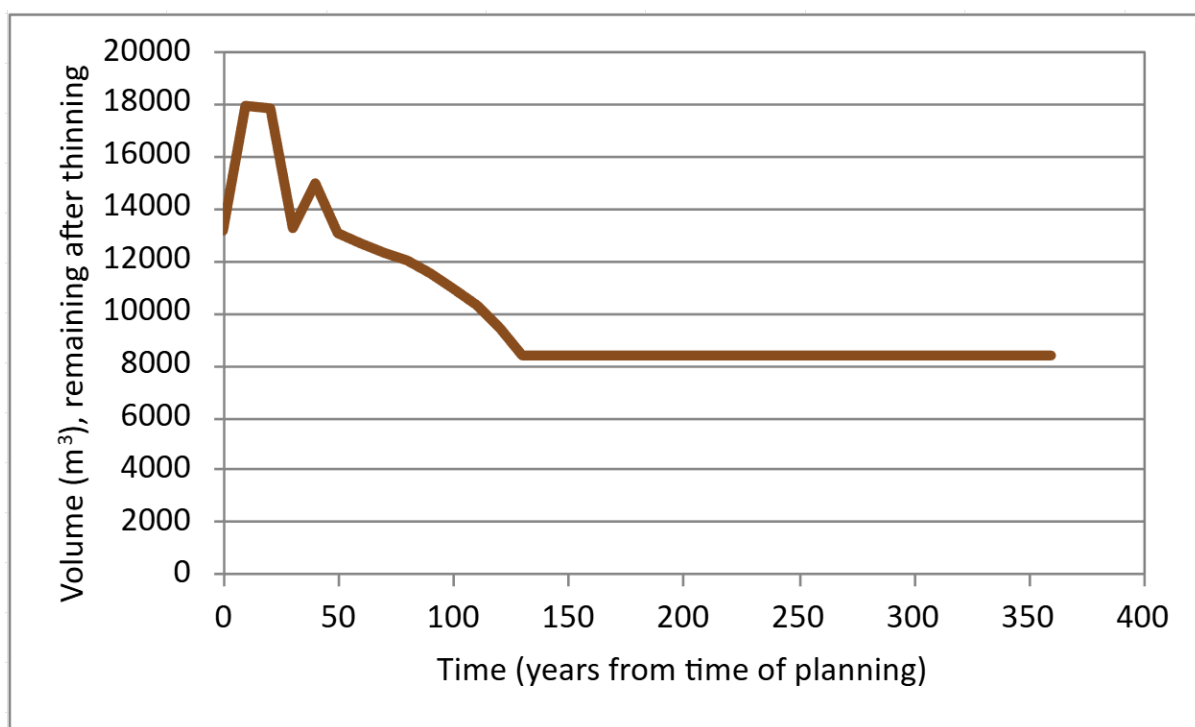
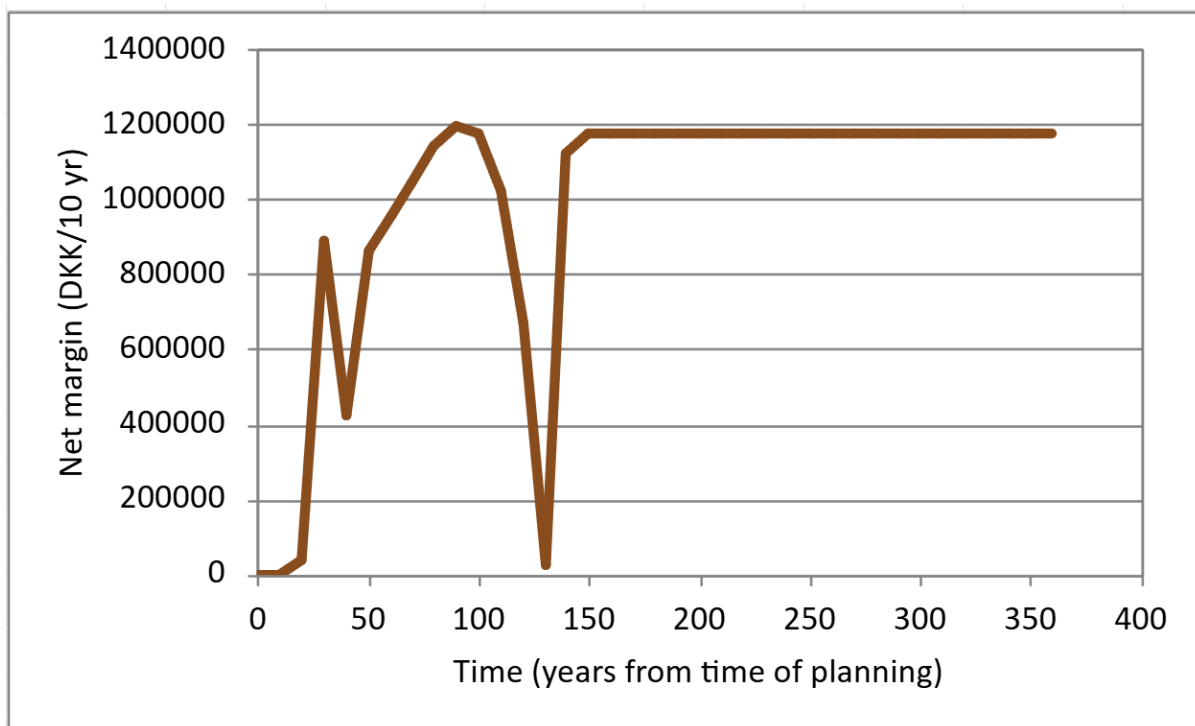
Timber Production Alternative				
Unit 1	FDT 14	Land Use Area (ha)		Age Average Species
Area Conifers	2,247	Beech (young)	8,713	55,56
		Beech (Old)	47,557	154,08
Area Broadleaf	58,963	Red alder	0,18	73,00
Age Average	86,6938776	Ash	1,38	59,00
		Oak	1,133	87,50
Total Area	69,81	Douglas Fir	0,79	85,00
Other uses	0,32	Norway Spruce	0,389	55,00
		Abies grandis	1,068	57,80
Unit 2	FDT 21	Land Use Area (ha)		Age Average Species
Area Conifers	6,6625	Douglas fir	1,453	32,25
Area Broadleaf	65,8915	Norway spruce	3,6115	89,86
		Oak	15,636	80,00
		Beech (Old)	12,069	144,79
Total Age Aver	79,4186047	Beech (young)	42,025	56,25
		Hornbeam	-	-
Total Area (ha)	86,83	Maple	5,2165	55,33
		Larch	0,877	105,50
		Cypress	0,234	-
		Ash	0,144	86,00
		Red alder	0,144	72,00
		Birch	0,684	85,00
		Scots Pine	0,487	74,33
		Lime	1,404	70,00
		Cherry	0,638	14,00

Figure B: Summary of existing tree species and their age for the future management units

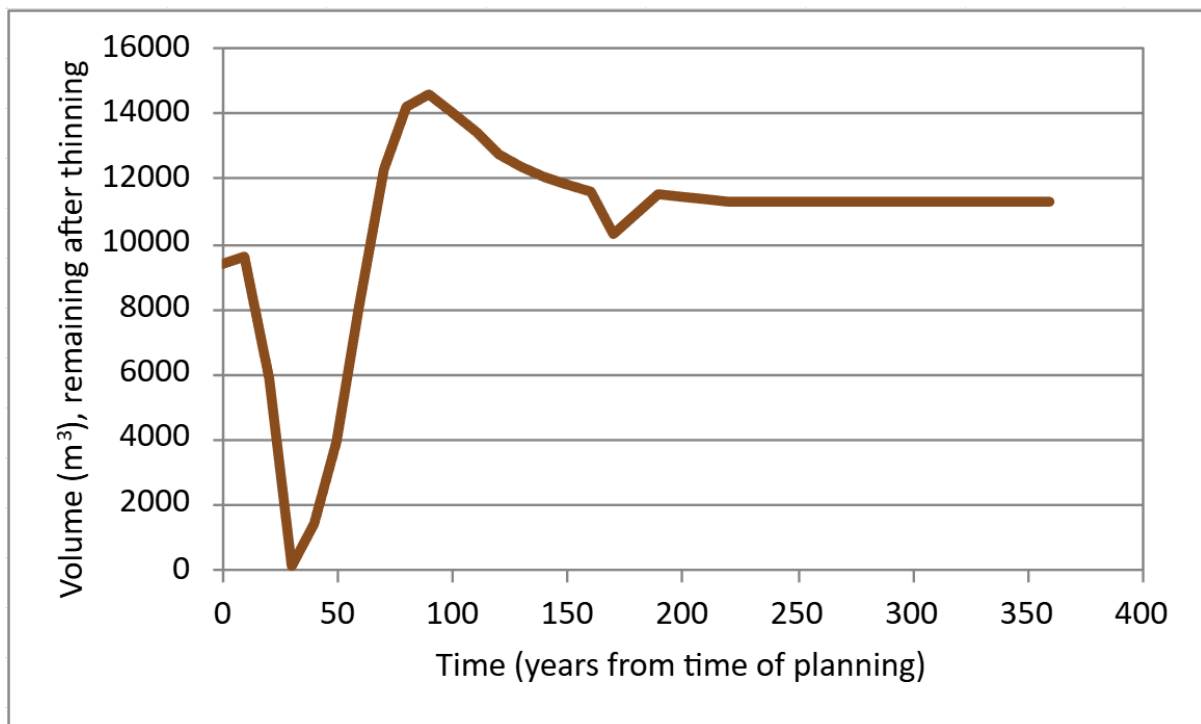
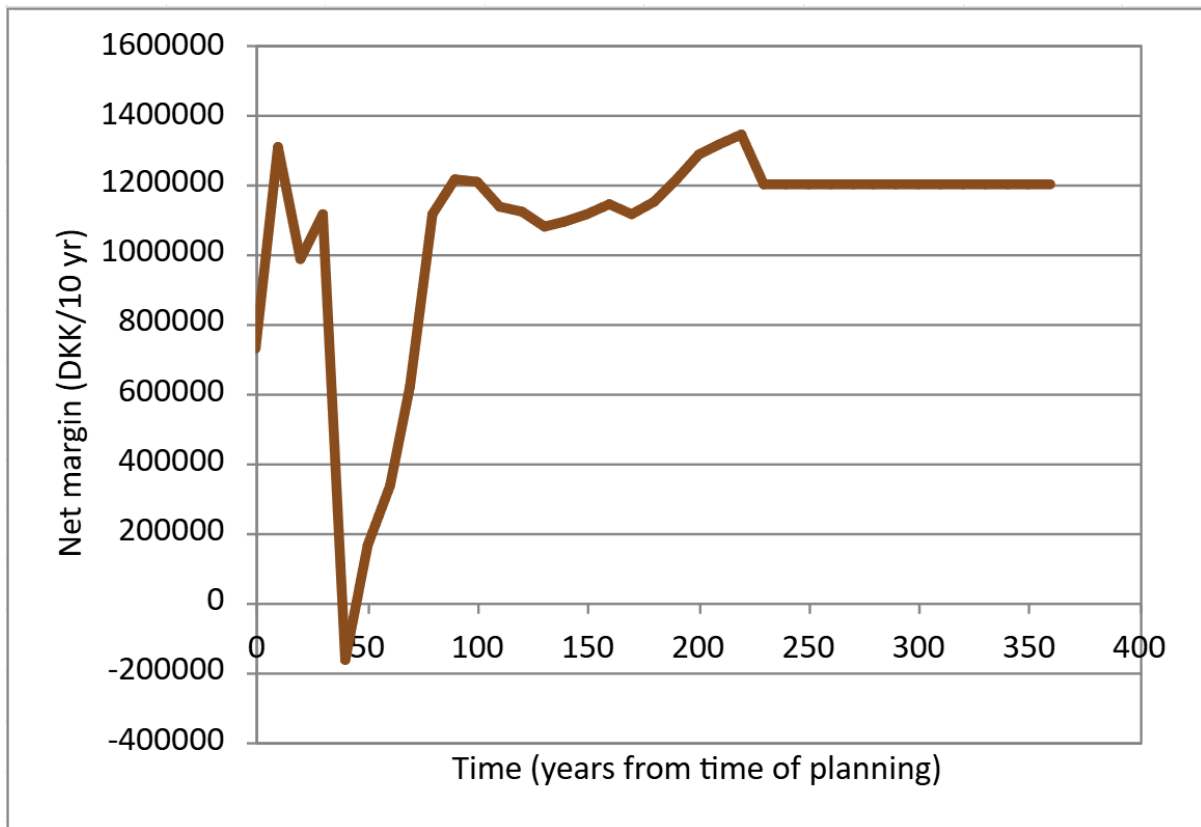
2. Economic Modelling



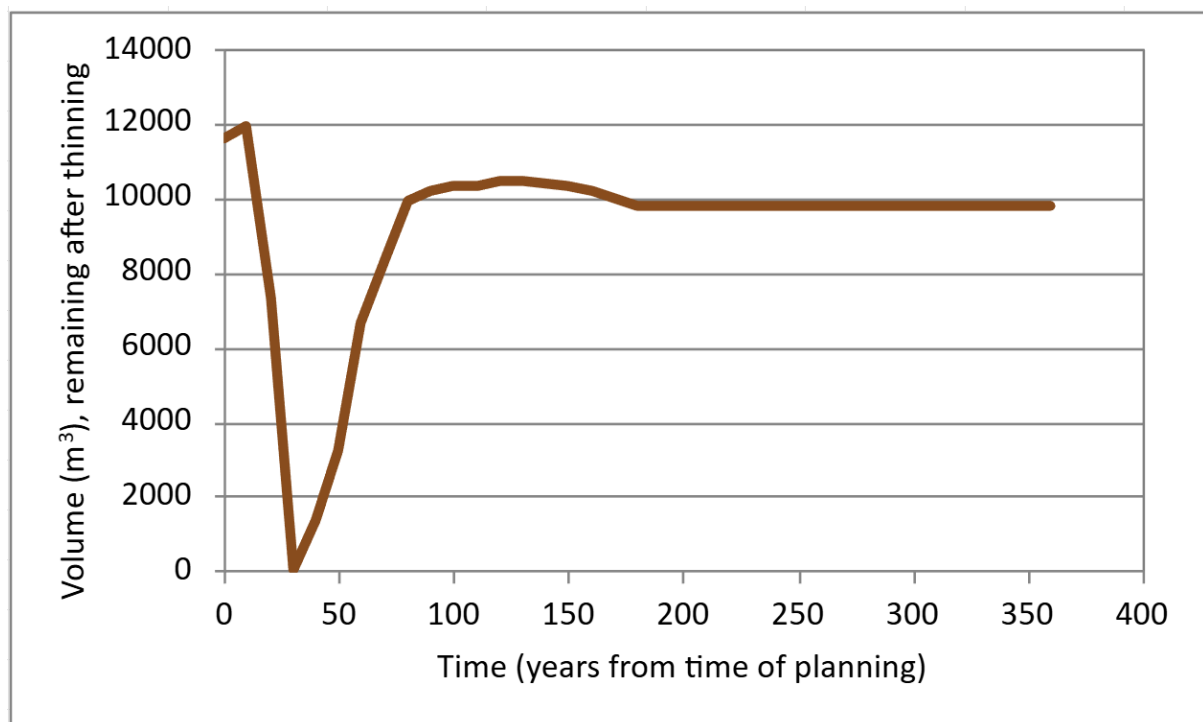
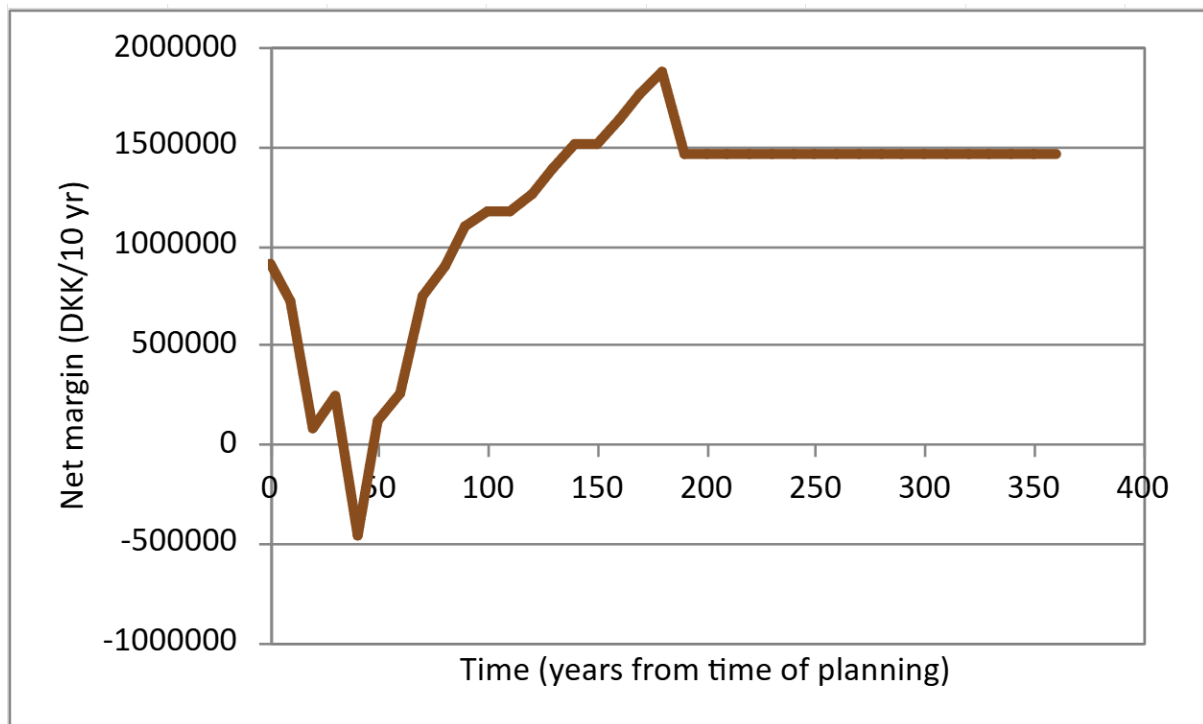
Figures C & D: Recreational Alternative: FDT 11



Figures E & F: Recreational Alternative: FDT 22



Figures G & H: Timber Production Alternative: FDT 14



Figures I & J: Timber Production Alternative: FDT 21

3. Carbon Modelling

Alternative FDT Area	Recreation		Combined		Timber Production		Combined		Biodiversity	
	FDT 11	FDT 22	133ha	Per ha	FDT 21	FDT 14	157 ha	Per ha	FDT 94	163
	63 ha	70 ha			87 ha	70 ha				
Total sequestration (Mg C (Area) ha-1 10 yr-1)	1569,76	1665,18	3234,94	24,32284511	102,66	82,48	185,14	1,17921019	3,446	21,1411043
Input to HWP pool (Mg C (Area) ha-1 10 yr-1)	200,31	0	200,31	1,506073026	666,27	535,64	1201,91	7,65544586	-	-
Total avoided emissions from energy substitution (Mg C (Area) ha-1 10 yr-1)	823,20	0	823,20	6,189498507	2738,16	2201,30	4939,46	31,461536	-	-
Total avoided emissions from products substitution (Mg C (Area) ha-1 10 yr-1)	185,08	0	185,08	1,391611476	615,63	494,93	1110,56	7,07363197	-	-
Total avoided emissions from all substitution (Mg C (Area) ha-1 10 yr-1)	1008,29	0	1008,29	7,581109983	3353,79	2696,23	6050,02	38,535168	-	-
Total carbon impact, without HWP (Mg C (Area) ha-1 10 yr-1)	2578,05	1665,18	4243,23	31,9039551	3456,45	2778,71	6235,16	39,7143782	3,446	21,1411043
Total carbon impact, with HWP input, but no HWP decay (Mg C (Area) ha-1 10 yr-1)	2778,35	1665,18	4443,53	33,41002812	4122,72	3314,35	7437,06	47,369824	3,446	21,1411043
Total carbon impact, without substitution (Mg C (Area) ha-1 10 yr-1)	1770,07	1665,18	3435,25	25,82891814	768,93	618,11	1387,04	8,83465605	3,446	21,1411043
Total carbon impact, with substitution (Mg C (Area) ha-1 10 yr-1)	2778,35	1665,18	4443,53	33,41002812	4122,72	3314,35	7437,06	47,369824	3,446	21,1411043

Table A: Results from the “Week_21_ForestCarbonManagement_Exercise_Data_Templates_May2025” spreadsheets

4. Hydrological Modelling

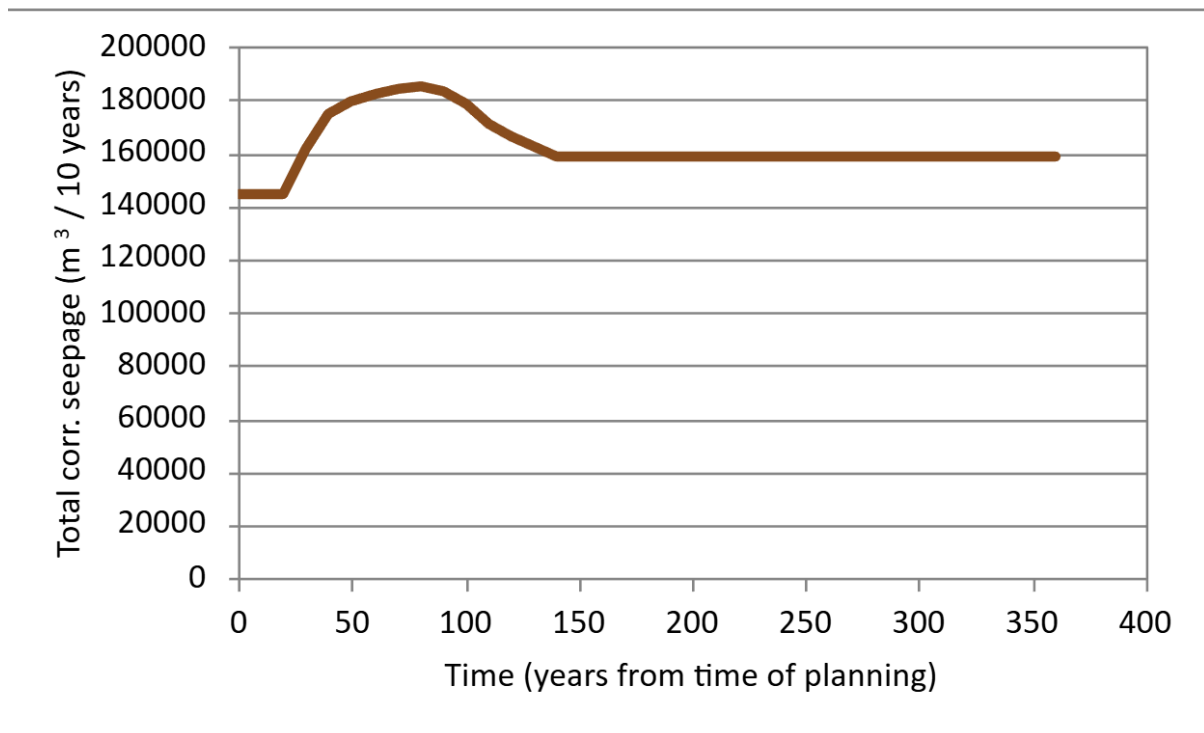


Figure K: Recreational Alternative: FDT 11

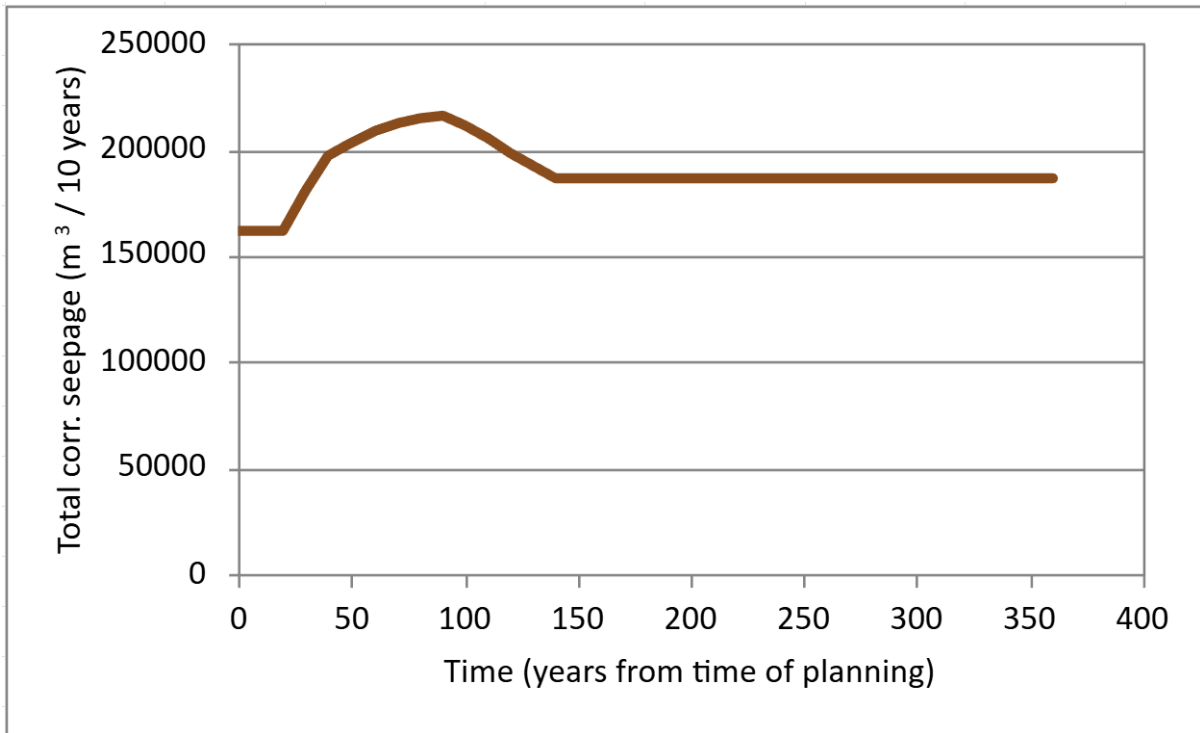


Figure L: Recreational Alternative: FDT 22

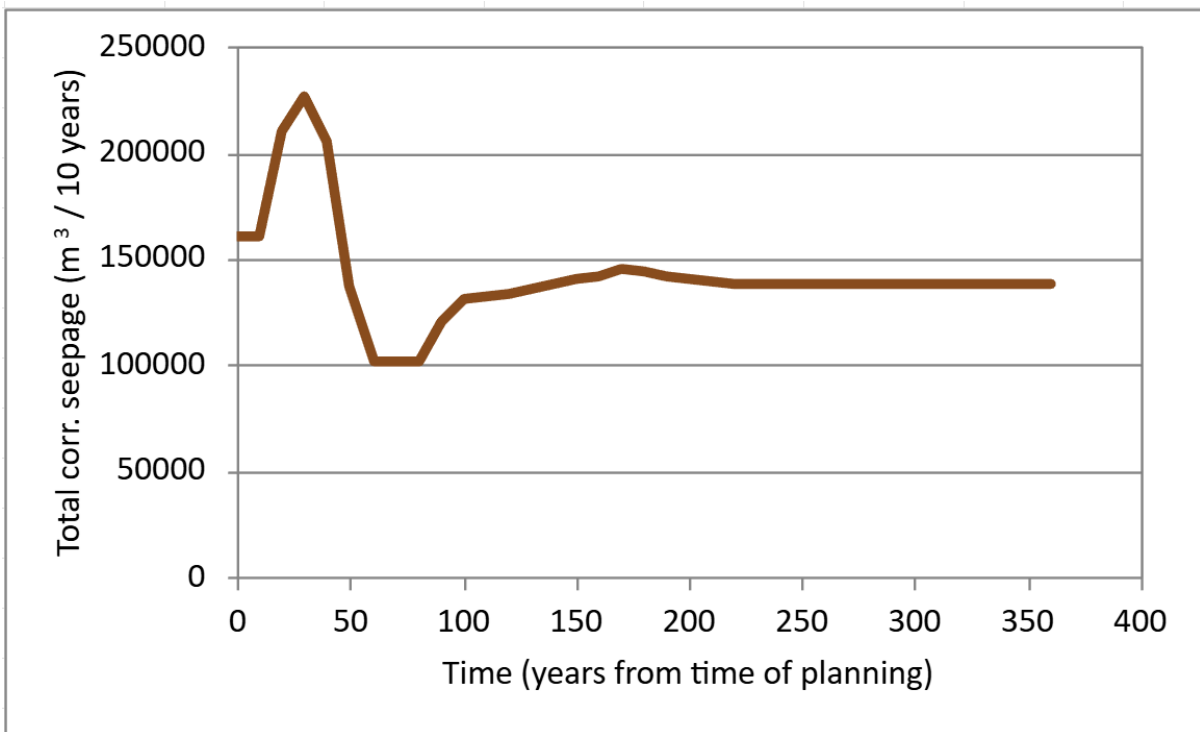


Figure M: Timber Production Alternative: FDT 14

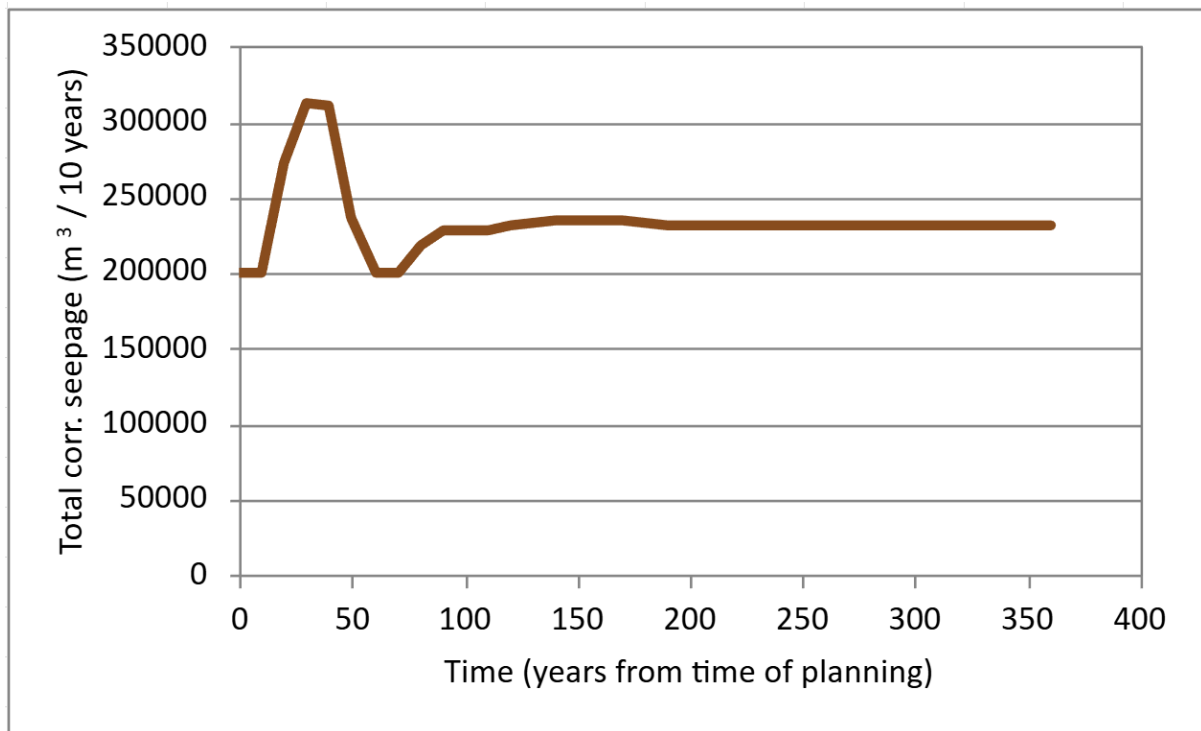


Figure N: Timber Production Alternative: FDT 21

	<i>EV in each FDT (DKK)</i>	<i>Total EV (DKK)</i>
Alternative 1 (Recreational focus)	FDT 22 = 1.601.160 FDT 11 = 1.385.749 FDT 94 = Unknown FDT 95 = Unknown	2.986.909
Alternative 2 (Timber Production focus)	FDT 14 = 1.362.270 FDT 21 = 1.980.048	3.342.318
Alternative 3 (Biodiversity focus)	FDT 94 = 3.504.335	3.504.335

Table B: Groundwater Economics

5. Biodiversity Modelling

Recreational Alternative					
Unit 1	FDT 22				
UNA Average					
1st Part	2nd Part	3rd Part	Total	Deadwood Only	
14,67	18,04	8,13	40,83	11,92	
Unit 2	FDT 11				
UNA Average					
1st Part	2nd Part	3rd Part	Total	Deadwood Only	
11,22	16,00	15,50	42,50	11,67	
Unit 3	FDT 94				
UNA Average					
1st Part	2nd Part	3rd Part	Total	Deadwood Only	
No available Data					14,00

Figure O: UNA Scores for existing Stands

Predicted UNA assessment for chosen management units in northern Jaegerborg Hegn

Compartment		Unit 1 (FDT 22)	Unit 2 (FDT 11)	Unit 3 (FDT 94)
Area (ha) (regarding the stand)		65,39	63	25
Investigator/Group:		Group 2	Group 2	Group 2
Date:		05.06.25	05.06.25	05.06.25
Start time:				
End time:				
No	Area			
1	Area > 1 ha	1	1	1
2	Area > 10 ha	1	1	1
3	Part of woodland with a coherent area > 100 ha	1	1	1
4	Bordering on other woodland or nature areas (e.g. wetland, lake, moorland, dunes, but not fields, urban areas or roads along at least 80% of the edge length.	0	1	1

5	Distance to cultivated fields, urban area, pig farms, larger roads >100 m for at least 95% of the area.	0	1	0
Stand structure				
6	Multi-layered stand structure (min. 3 layers) in at least 10% of the tree covered area	1	1	1
7	Patchy height variation in the stand on at least 10% of the tree covered area (min. 3 patches of each 400-1000 m ² /ha)	1	1	1
8	Natural, small gaps (> 100 m ²). E.g. wetlands, storm gaps.	1	1	1
9	No visible signs of planting of the stand. E.g. no sign of planting rows etc. (< 5% of planted trees is acceptable)	1	1	1
Tree species				
10	Fagus, Quercus robur or Q. sessiliflora; dbh ≥ 10 cm	1	1	1
11	Fraxinus; dbh ≥ 10 cm	0	0	1
12	Alnus glutinosa; dbh ≥ 10 cm	0	0	1
13	Tilia cordata; dbh ≥ 10 cm	1	0	1
14	Tilia cordata; dbh ≥ 25 cm	1	0	1
15	Carpinus; dbh ≥ 10 cm	1	0	1
16	Ulmus; dbh ≥ 10 cm	1	0	0

17	Acer campestre, A. platanoides; dbh $\geq$ 10 cm	0	0	1
18	Populus tremula, Betula, Sorbus, Prunus avium, Malus silvestris or Corylus; dbh $\geq$ 10 cm; Ilex; dbh > 5 cm.	0	0	0
19	No larger occurrence of Prunus serotina, Picea sitchensis nor other invasive tree species nor occurrence of giant hogweed (Hieracleum mantegazzianum) . Trees; independent of species	1	1	1
20	Large trees; dbh $\geq$ 50 cm present	1	1	1
21	Large trees; dbh $\geq$ 50 cm $\geq$ 5/ha	1	1	1
22	Large trees; dbh $\geq$ 50 cm $\geq$ 10/ha	1	1	1
23	Large trees; dbh $\geq$ 75cm present	1	1	1
24	Large trees; dbh $\geq$ 75cm $\geq$ 1/ha	1	1	1
25	Large trees; dbh $\geq$ 75cm $\geq$ 5/ha	1	1	1
26	Megatrees (dbh $\geq$ 100 cm) present	1	1	1
27	Megatrees (dbh $\geq$ 100 cm); $\geq$ 1/ha	1	1	1
28	Megatrees (dbh $\geq$ 100 cm); $\geq$ 5/ha	1	1	1
29	Living trees with larger holes, hollows, wounds etc.; dbh >25 cm; $\geq$ 1 tree/ha	1	1	1
30	Living trees with larger holes, hollows, wounds	1	1	1

	etc.; dbh >50 cm, ≥1 tree/ha			
31	Fallen or hanging trees, still living, dbh ≥25	1	1	1
32	Fallen or hanging trees, still living, dbh ≥50	1	1	1
PART SUM 1		26	24	29
Canopy (the upper crown layer, independent of tree/stand height)				
33	Large diameter variation of trees in canopy (>50 cm i dbh)	1	1	1
34	Large age variation of trees in canopy (> 100 years)	1	1	1
35	Large shape variation among canopy trees	1	1	1
36	At least 4 tree species in canopy	1	1	1
37	At least 3 domestic species with at least 25% of canopy	1	1	1
Understorey/reg eneration (woody plants > ½ m og < 5 m)				
38	Understorey/reg eneration (woody plants incl. Corylus (Hazel), but not bushes) present on 10-35 % of the area.	0	0	1
39	Regeneration unevenaged; age variation >50 years	1	1	1
40	Regeneration in patches	1	1	1

41	≥1 domestic tree species in understorey/regeneration	1	1	1
42	>5 domestic tree species in understorey/regeneration	0	1	1
43	≥10 domestic tree species in understorey/regeneration	0	0	1
44	>5 domestic bush species in understorey	0	0	1
45	≥10 domestic bush species in understorey	0	0	1
Dead wood; Non-sawn trees, independent of species dbh= diameter at breast height (1,3 m) in upright position				
46	Dead or dying trees, dbh 10-24 cm	1	1	1
47	Dead or dying trees, dbh 25-49 cm	1	1	1
48	Dead or dying trees, dbh 50-74 cm	1	1	1
49	Dead or dying trees, dbh 75- 99 cm	0	0	1
50	Dead or dying trees, dbh ≥100 cm (megatrees)	0	0	1
51	Dead trees, dbh ≥25 cm, several (≥5/ha)	1	1	1
52	Dead trees, dbh ≥25 cm, many (>10/ha)	1	1	1
53	Dead trees, dbh ≥50 cm; many (≥5/ha)	1	1	1

54	Dead trees, dbh ≥50 cm; >10/ha	1	1	1
55	Dead trees, dbh ≥75 cm; ≥5/ha	1	1	1
56	Dead trees, dbh ≥75 cm; >10/ha	1	1	1
57	Standing dead trees; dbh 10-24 cm	1	1	1
58	Standing dead trees; dbh 25-49 cm	1	1	1
59	Standing dead trees; dbh ≥50 cm	1	1	1
60	Dead uprooted, fallen trees, dbh ≥25cm	1	1	1
61	Dead uprooted, fallen trees, dbh ≥50 cm	1	1	1
62	Dead uprooted, fallen trees, dbh ≥75 cm	1	1	1
63	Dead uprooted, fallen trees, dbh ≥100 cm	0	0	1
64	Dead logs, exposed to direct sunlight dbh ≥25 cm	0	0	0
65	Dead logs, exposed to direct sunlight dbh ≥50 cm	0	0	0
66	Living or dead snags; height > 2m, dbh >25 cm	1	1	1
67	Living or dead snags; height > 2m, dbh >50 cm	1	1	1
68	Dead trees dbh ≥25 cm, at least 3 species	1	1	1
69	Dead trees dbh ≥25 cm, at least 5 species	1	1	1
70	Dead logs dbh ≥25 cm, Decay class 1 (Fresh,	1	1	1

	newly dead, wood hard, bark intact. Unchanged (round) shape in cross-section.			
71	Dead logs dbh ≥25 cm, Decay class 2 (Soft at surface (to approx. 1 cm depth); bark loose or partly fallen off; unchanged (round) shape in cross-section).	1	1	1
72	Dead logs dbh ≥25 cm, Decay class 3 (soft several cm in depth. Bark lost (except tree species decaying inside the bark (e.g. Ilex and Betula)).	1	1	1
73	Dead logs dbh ≥25 cm, Decay class 4 (log rotten, with holes, breaks easily in parts. Oval in cross- section.)	1	1	1
74	Dead logs dbh ≥25 cm, Decay class 5 (log wood more or less decomposed; visible as rough fragments or as different vegetation.)	1	1	1
	PART SUM 2	32	33	40
Flora				
75	Forest floor vegetation of summergreen vascular plants (woodland	0	1	1

	species) present at at least 10% of the area.			
76	Living stems (boles) rich in bryophytes (mosses). > 25% coverage up to 3 meters.	0	0	1
77	Living stems (boles) rich in lichens (arbuscular or thallose). > 25% coverage up to 3 meters.	1	1	1
Topography and soil				
78	Large macrotopographic variation (>20 meters within 1 ha)	0	0	0
79	Large microtopographic variation (>1m/100m ²)	1	1	1
80	Large mounds from storm-felled trees present	1	1	1
81	Large stones, rocks; min. 1 x 1 meter; natural occurrences in the surface.	0	0	0
82	Mull present on at least 5% of the area	1	1	1
83	Mor layers present on at least 5% of the area	1	1	1
84	Raw humus deposits (mor or peat), layers ≥5 cm. On at least 5% of the area	0	0	0
85	Chalk visible at the surface.	0	0	0

86	Patches with sun exposed, naturally exposed soil surface (landslide, dune etc.)	0	0	0
Water level, hydrology				
87	Open, treeless wetlands; min 5% of area.	0	0	0
88	Swamp forest present (eg. Alderswamp, birch swampforest, mixed swamp forest); min 5% of area	0	0	0
89	Temporary waterlogged areas (temporary lakes and ponds).	0	0	0
90	Humid hollows present (size min 100 m2)	0	0	0
91	Wells present	0	0	0
92	Natural, unregulated watercourses	0	0	1
93	Wetlands without ditches or ditches efficiently closed	0	0	0
94	Wet or humid soil prevailing on min. 25% of the area	0	0	0
Management impacts				
95	No signs of soil cultivation	1	1	1
96	No ditches or ditches closed efficiently	1	1	1
97	No newly managed ditches (within the last 10 years)	1	1	1

98	No tracks from motor vehicles, driving in the stand (deep wheel tracks)	1	1	1
99	No signs of cutting (No stumps)	0	0	1
100	No younger signs of cutting (No stumps younger than 10 year)	0	0	1
	PART SUM 3	9	10	14
	Part sum 1 (transfer)	26	24	29
	Part sum 2 (transfer)	32	33	40
TOTAL SUM (Index value)		67	67	83

Table C: Predicted UNA Index for future stands

Biodiversity Alternative:

Week 19, Predicted UNA assessment for chosen management units in northern Jaegerborg Hegn

Compartment		FDT 94 (Deciduous)	FDT 94 (Conifers)
Area (ha) (regarding the stand)		152,16	153,16
Investigator/Group:		Group 2	Group 3
Date:		05.06.25	06.06.25
Start time:			
End time:			
No	Area		
1	Area > 1 ha	1	1
2	Area >10 ha	1	1
3	Part of woodland with a coherent area > 100 ha	1	1
4	Bordering on other woodland or nature areas (e.g. wetland, lake, moorland, dunes, but not fields, urban areas or roads along at least 80% of the edge length.	1	1

5	Distance to cultivated fields, urban area, pig farms, larger roads >100 m for at least 95% of the area.	0	0
Stand structure			
6	Multi-layered stand structure (min. 3 layers) in at least 10% of the tree covered area	1	0
7	Patchy height variation in the stand on at least 10% of the tree covered area (min. 3 patches of each 400-1000 m ² /ha)	1	1
8	Natural, small gaps (> 100 m ²). E.g. wetlands, storm gaps.	1	1
9	No visible signs of planting of the stand. E.g. no sign of planting rows etc. (< 5% of planted trees is acceptable)	1	1
Tree species			
10	Fagus, Quercus robur or Q. sessiliflora; dbh ≥ 10 cm	1	0
11	Fraxinus; dbh ≥ 10 cm	1	0
12	Alnus glutinosa; dbh ≥ 10 cm	1	0
13	Tilia cordata; dbh ≥ 10 cm	1	0
14	Tilia cordata; dbh ≥ 25 cm	1	0
15	Carpinus; dbh ≥ 10 cm	1	0
16	Ulmus; dbh ≥ 10 cm	0	0
17	Acer campestre, A. platanooides; dbh ≥ 10 cm	1	0
18	Populus tremula, Betula, Sorbus, Prunus avium, Malus silvestris or Corylus; dbh ≥ 10 cm; Ilex; dbh > 5 cm.	0	0

19	No larger occurrence of <i>Prunus serotina</i> , <i>Picea sitchensis</i> nor other invasive tree species nor occurrence of giant hogweed (<i>Hieracleum mantegazzianum</i>). Trees; independent of species	1	1
20	Large trees; dbh ≥ 50 cm present	1	1
21	Large trees; dbh ≥ 50 cm ≥ 5 /ha	1	1
22	Large trees; dbh ≥ 50 cm ≥ 10 /ha	1	1
23	Large trees; dbh ≥ 75 cm present	1	1
24	Large trees; dbh ≥ 75 cm ≥ 1 /ha	1	1
25	Large trees; dbh ≥ 75 cm ≥ 5 /ha	1	1
26	Megatrees (dbh ≥ 100 cm) present	1	1
27	Megatrees (dbh ≥ 100 cm); ≥ 1 /ha	1	1
28	Megatrees (dbh ≥ 100 cm); ≥ 5 /ha	1	0
29	Living trees with larger holes, hollows, wounds etc.; dbh > 25 cm; ≥ 1 tree/ha	1	1
30	Living trees with larger holes, hollows, wounds etc.; dbh > 50 cm, ≥ 1 tree/ha	1	1
31	Fallen or hanging trees, still living, dbh ≥ 25	1	1
32	Fallen or hanging trees, still living, dbh ≥ 50	1	1
PART SUM 1		29	20
Canopy (the upper crown layer, independent of tree/stand height)			
33	Large diameter variation of trees in canopy (> 50 cm i dbh)	1	0

34	Large age variation of trees in canopy (> 100 years)	1	0
35	Large shape variation among canopy trees	1	0
36	At least 4 tree species in canopy	1	1
37	At least 3 domestic species with at least 25% of canopy	1	0
Understorey/regeneration (woody plants > ½ m og < 5 m)			
38	Understorey/regeneration (woody plants incl. Corylus (Hazel), but not bushes) present on 10-35 % of the area.	1	0
39	Regeneration unevenaged; age variation >50 years	1	1
40	Regeneration in patches	1	1
41	≥1 domestic tree species in understorey/regeneration	1	1
42	>5 domestic tree species in understorey/regeneration	1	0
43	≥10 domestic tree species in understorey/regeneration	1	1
44	>5 domestic bush species in understorey	1	0
45	≥10 domestic bush species in understorey	1	0
Dead wood; Non-sawn trees, independent of species dbh= diameter at breast height (1,3 m) in upright position			
46	Dead or dying trees, dbh 10-24 cm	1	1

47	Dead or dying trees, dbh 25-49 cm	1	1
48	Dead or dying trees, dbh 50-74 cm	1	1
49	Dead or dying trees, dbh 75- 99 cm	1	1
50	Dead or dying trees, dbh ≥ 100 cm (megatrees)	1	1
51	Dead trees, dbh ≥ 25 cm, several (≥ 5 /ha)	1	0
52	Dead trees, dbh ≥ 25 cm, many (>10 /ha)	1	1
53	Dead trees, dbh ≥ 50 cm; many (≥ 5 /ha)	1	1
54	Dead trees, dbh ≥ 50 cm; >10 /ha	1	1
55	Dead trees, dbh ≥ 75 cm; ≥ 5 /ha	1	0
56	Dead trees, dbh ≥ 75 cm; >10 /ha	1	0
57	Standing dead trees; dbh 10-24 cm	1	1
58	Standing dead trees; dbh 25-49 cm	1	1
59	Standing dead trees; dbh ≥ 50 cm	1	1
60	Dead uprooted, fallen trees, dbh ≥ 25 cm	1	1
61	Dead uprooted, fallen trees, dbh ≥ 50 cm	1	1
62	Dead uprooted, fallen trees, dbh ≥ 75 cm	1	0
63	Dead uprooted, fallen trees, dbh ≥ 100 cm	1	0
64	Dead logs, exposed to direct sunlight dbh ≥ 25 cm	1	1
65	Dead logs, exposed to direct sunlight dbh ≥ 50 cm	1	1
66	Living or dead snags; height > 2 m, dbh > 25 cm	1	1
67	Living or dead snags; height > 2 m, dbh > 50 cm	1	1
68	Dead trees dbh ≥ 25 cm, at least 3 species	1	1

69	Dead trees dbh ≥ 25 cm, at least 5 species	1	1
70	Dead logs dbh ≥ 25 cm, Decay class 1 (Fresh, newly dead, wood hard, bark intact. Unchanged (round) shape in cross-section).	1	1
71	Dead logs dbh ≥ 25 cm, Decay class 2 (Soft at surface (to approx. 1 cm depth); bark loose or partly fallen off; unchanged (round) shape in cross-section).	1	1
72	Dead logs dbh ≥ 25 cm, Decay class 3 (soft several cm in depth. Bark lost (except tree species decaying inside the bark (e.g. Ilex and Betula)).	1	1
73	Dead logs dbh ≥ 25 cm, Decay class 4 (log rotten, with holes, breaks easily in parts. Oval in cross-section.)	1	1
74	Dead logs dbh ≥ 25 cm, Decay class 5 (log wood more or less decomposed; visible as rough fragments or as different vegetation.)	1	1
PART SUM 2		29	24
Flora			
75	Forest floor vegetation of summergreen vascular plants (woodland species) present at at least 10% of the area.	1	1
76	Living stems (boles) rich in bryophytes (mosses). > 25% coverage up to 3 meters.	1	0

77	Living stems (boles) rich in lichens (arbuscular or thallose). > 25% coverage up to 3 meters.	1	0
Topography and soil			
78	Large macrotopographic variation (>20 meters within 1 ha)	0	0
79	Large microtopographic variation (>1m/100m ²)	1	1
80	Large mounds from storm-felled trees present	1	1
81	Large stones, rocks; min. 1 x 1 meter; natural occurrences in the surface.	1	1
82	Mull present on at least 5% of the area	1	1
83	Mor layers present on at least 5% of the area	1	1
84	Raw humus deposits (mor or peat), layers ≥5 cm. On at least 5% of the area	1	1
85	Chalk visible at the surface.	0	0
86	Patches with sun exposed, naturally exposed soil surface (landslide, dune etc.)	1	1
Water level, hydrology			
87	Open, treeless wetlands; min 5% of area.	1	1
88	Swamp forest present (eg. Alderswamp, birch swampforest, mixed swamp forest); min 5% of area	1	1
89	Temporary waterlogged areas	1	1

	(temporary lakes and ponds).		
90	Humid hollows present (size min 100 m2)	1	1
91	Wells present	0	0
92	Natural, unregulated watercourses	1	1
93	Wetlands without ditches or ditches efficiently closed	1	1
94	Wet or humid soil prevailing on min. 25% of the area	1	1
Management impacts			
95	No signs of soil cultivation	1	1
96	No ditches or ditches closed efficiently	1	1
97	No newly managed ditches (within the last 10 years)	1	1
98	No tracks from motor vehicles, driving in the stand (deep wheel tracks)	1	1
99	No signs of cutting (No stumps)	1	1
100	No younger signs of cutting (No stumps younger than 10 year)	1	1
	PART SUM 3	23	21
	Part sum 1 (transfer)	29	20
	Part sum 2 (transfer)	29	24
TOTAL SUM (Index value)		81	65

Table D: Predicted UNA Index for future stands

Timber production alternative

Figure x: UNA Scores for existing Stands

Week 19, Predicted UNA assessment for chosen management units in northern Jaegerborg Hegn

Compartment		Unit 1 (FDT 14)	Unit 2 (FDT 21)
Area (ha) (regarding the stand)		70	87
Investigator/Group:		Group 2	Group 2
Date:		05.06.25	05.06.25
Start time:			
End time:			
No	Area		
1	Area > 1 ha	1	1
2	Area > 10 ha	1	1
3	Part of woodland with a coherent area > 100 ha	1	1
4	Bordering on other woodland or nature areas (e.g. wetland, lake, moorland, dunes, but not fields, urban areas or roads along at least 80% of the edge length.	1	1
5	Distance to cultivated fields, urban area, pig farms, larger roads > 100 m for at least 95% of the area.	0	0
Stand structure			
6	Multi-layered stand structure (min. 3 layers) in at least 10% of the tree covered area	1	1
7	Patchy height variation in the stand on at least 10% of the tree covered area (min. 3 patches of each 400-1000 m ² /ha)	1	1
8	Natural, small gaps (> 100 m ²). E.g. wetlands, storm gaps.	1	1
9	No visible signs of planting of the stand. E.g. no sign of planting rows etc. (< 5% of planted trees is acceptable)	0	0
Tree species			

10	Fagus, Quercus robur or Q. sessiliflora; dbh ≥ 10 cm	1	1
11	Fraxinus; dbh ≥ 10 cm	1	1
12	Alnus glutinosa; dbh ≥10 cm	0	0
13	Tilia cordata; dbh ≥10 cm	0	1
14	Tilia cordata; dbh ≥25 cm	0	1
15	Carpinus; dbh ≥ 10 cm	0	1
16	Ulmus; dbh ≥ 10 cm	0	0
17	Acer campestre, A. platanoides; dbh ≥ 10 cm	0	0
18	Populus tremula, Betula, Sorbus, Prunus avium, Malus silvestris or Corylus; dbh ≥ 10 cm; Ilex; dbh > 5 cm.	0	0
19	No larger occurrence of Prunus serotina, Picea sitchensis nor other invasive tree species nor occurrence of giant hogweed (Hieracleum mantegazzianum). Trees; independent of species	0	0
20	Large trees; dbh ≥50 cm present	1	1
21	Large trees; dbh ≥50 cm ≥5/ha	1	1
22	Large trees; dbh ≥50 cm ≥10/ha	1	1
23	Large trees; dbh ≥75cm present	0	0
24	Large trees; dbh ≥75cm ≥1/ha	0	0
25	Large trees; dbh ≥75cm ≥5/ha	0	0
26	Megatrees (dbh ≥ 100 cm) present	0	0
27	Megatrees (dbh ≥ 100 cm); ≥1/ha	0	0
28	Megatrees (dbh ≥ 100 cm); ≥5/ha	0	0

29	Living trees with larger holes, hollows, wounds etc.; dbh >25 cm; ≥1 tree/ha	1	1
30	Living trees with larger holes, hollows, wounds etc.; dbh >50 cm, ≥1 tree/ha	1	1
31	Fallen or hanging trees, still living, dbh ≥25	1	1
32	Fallen or hanging trees, still living, dbh ≥50	1	1
PART SUM 1		16	19
Canopy (the upper crown layer, independent of tree/stand height)			
33	Large diameter variation of trees in canopy (>50 cm i dbh)	1	1
34	Large age variation of trees in canopy (> 100 years)	1	1
35	Large shape variation among canopy trees	0	0
36	At least 4 tree species in canopy	1	1
37	At least 3 domestic species with at least 25% af canopy	1	1
Understorey/regeneration (woody plants > ½ m og < 5 m)			
38	Understorey/regeneration (woody plants incl. Corylus (Hazel), but not bushes) present on 10-35 % of the area.	0	0
39	Regeneration unevenaged; age variation >50 years	1	1
40	Regeneration in patches	1	1
41	≥1 domestic tree species in understorey/regeneration	1	1

42	>5 domestic tree species in understorey/regeneration	1	1
43	≥10 domestic tree species in understorey/regeneration	0	0
44	>5 domestic bush species in understorey	0	0
45	≥10 domestic bush species in understorey	0	0
Dead wood; Non-sawn trees, independent of species dbh= diameter at breast height (1,3 m) in upright position			
46	Dead or dying trees, dbh 10-24 cm	1	1
47	Dead or dying trees, dbh 25-49 cm	1	1
48	Dead or dying trees, dbh 50-74 cm	0	0
49	Dead or dying trees, dbh 75- 99 cm	0	0
50	Dead or dying trees, dbh ≥100 cm (megatrees)	0	0
51	Dead trees, dbh ≥25 cm, several (≥5/ha)	1	1
52	Dead trees, dbh ≥25 cm, many (>10/ha)	0	0
53	Dead trees, dbh ≥50 cm; many (≥5/ha)	1	1
54	Dead trees, dbh ≥50 cm; >10/ha	1	1
55	Dead trees, dbh ≥75 cm; ≥5/ha	0	0
56	Dead trees, dbh ≥75 cm; >10/ha	0	0
57	Standing dead trees; dbh 10-24 cm	1	1
58	Standing dead trees; dbh 25-49 cm	1	1
59	Standing dead trees; dbh ≥50 cm	1	1

60	Dead uprooted, fallen trees, dbh ≥ 25 cm	1	1
61	Dead uprooted, fallen trees, dbh ≥ 50 cm	1	1
62	Dead uprooted, fallen trees, dbh ≥ 75 cm	0	0
63	Dead uprooted, fallen trees, dbh ≥ 100 cm	0	0
64	Dead logs, exposed to direct sunlight dbh ≥ 25 cm	0	0
65	Dead logs, exposed to direct sunlight dbh ≥ 50 cm	0	0
66	Living or dead snags; height > 2m, dbh >25 cm	1	1
67	Living or dead snags; height > 2m, dbh >50 cm	1	1
68	Dead trees dbh ≥ 25 cm, at least 3 species	1	1
69	Dead trees dbh ≥ 25 cm, at least 5 species	1	1
70	Dead logs dbh ≥ 25 cm, Decay class 1 (Fresh, newly dead, wood hard, bark intact. Unchanged (round) shape in cross-section.	1	1
71	Dead logs dbh ≥ 25 cm, Decay class 2 (Soft at surface (to approx. 1 cm depth); bark loose or partly fallen off; unchanged (round) shape in cross-section).	1	1
72	Dead logs dbh ≥ 25 cm, Decay class 3 (soft several cm in depth. Bark lost (except tree species decaying inside the bark (e.g. Ilex and Betula)).	1	1
73	Dead logs dbh ≥ 25 cm, Decay class 4 (log rotten, with holes,	1	1

	breaks easily in parts. Oval in cross-section.)		
74	Dead logs dbh ≥ 25 cm, Decay class 5 (log wood more or less decomposed; visible as rough fragments or as different vegetation.)	1	1
	PART SUM 2	27	27
Flora			
75	Forest floor vegetation of summergreen vascular plants (woodland species) present at at least 10% of the area.	1	1
76	Living stems (boles) rich in bryophytes (mosses). > 25% coverage up to 3 meters.	0	0
77	Living stems (boles) rich in lichens (arbuscular or thallose). > 25% coverage up to 3 meters.	0	0
Topography and soil			
78	Large macrotopographic variation (>20 meters within 1 ha)	0	0
79	Large microtopographic variation (>1m/100m ²)	1	1
80	Large mounds from storm-felled trees present	0	0
81	Large stones, rocks; min. 1 x 1 meter; natural occurrences in the surface.	0	0
82	Mull present on at least 5% of the area	0	0
83	Mor layers present on at least 5% of the area	1	1

84	Raw humus deposits (mor or peat), layers ≥ 5 cm. On at least 5% of the area	0	0
85	Chalk visible at the surface.	0	0
86	Patches with sun exposed, naturally exposed soil surface (landslide, dune etc.)	0	0
Water level, hydrology			
87	Open, treeless wetlands; min 5% of area.	0	0
88	Swamp forest present (eg. Alderswamp, birch swampforest, mixed swamp forest); min 5% of area	0	0
89	Temporary waterlogged areas (temporary lakes and ponds).	0	0
90	Humid hollows present (size min 100 m ²)	0	0
91	Wells present	0	0
92	Natural, unregulated watercourses	0	0
93	Wetlands without ditches or ditches efficiently closed	0	0
94	Wet or humid soil prevailing on min. 25% of the area	0	0
Management impacts			
95	No signs of soil cultivation	0	0
96	No ditches or ditches closed efficiently	1	1
97	No newly managed ditches (within the last 10 years)	1	1
98	No tracks from motor vehicles, driving in the stand (deep wheel tracks)	0	0
99	No signs of cutting (No stumps)	0	0

100	No younger signs of cutting (No stumps younger than 10 year)	0	0
	PART SUM 3	5	5
	Part sum 1 (transfer)	16	19
	Part sum 2 (transfer)	27	27
TOTAL SUM (Index value)		48	51

Table E: Predicted UNA Index for future stands

Figure P: Natura2000 habitats at Bøllemosen and their appropriate FDT. Source: (Larsen, Jakobsen, Jensen, Nielsen, & Granat, 2005)

Habitat code	Site description	FDT
3160	Natural dystrophic lakes and ponds	Not specified
7140	Transition mires and quaking bogs	Not specified
9110	Luzulo-Fagetum beech forests	FDT 11 (Beech)
9160	Sub-Atlantic and medio-European oak or oak-hornbeam forests of the Carpinion betuli	FDT 21 (Oak with ash and hornbeam)
91D0	Bog Woodlands	41 (Birch with Scots pine and spruce)

Figure Q: Suitable FDTs per soil classification in northern Jægersborg Hegn.

Soil classification found in northern Jægersborg Hegn	FDT's suitable in soil condition
33	11, 13, 14, 22, 51, 52, 61, 71, 91, 92, 93, 94
44	11, 12, 13, 14, 61, 71, 91, 92, 93, 94
55	11, 12, 13, 14, 21, 91, 92, 93, 94
66	12, 21, 91, 92, 93, 94
74	12, 21, 91
83	21
92t	(41)*

93t	(31, 41)*
94t	(31)*
94	31
95	31

**Peat soil*

Figure R: Suitable FDTs per management goal.

Recreation	Production	Biological
11, 12, (21), 22, 23, (52), (61), 82, 92, 93, 94	11, 12, 13, 14, 21, 51, 52, 61, 71, (91)	12, (21), 22, 23, 31, 41, 81, 82, 91, 92, 93, 94

Orchard Meadow

Forest development type 95



Species Distribution

Fruit Trees:

- Apple: 40%
- Pear: 25%
- Plum: 20%
- Cherry: 10%
- Miscellaneous (e.g., Walnut, Medlar): 5%

Understory Vegetation:

- Grasses (e.g., Fescues, Ryegrass): 35%
- Wildflowers (e.g., Clover, Dandelion, Yarrow): 40%
- Herbs and Shrubs (e.g., Thyme, Wild Rose): 25%

Stand Structure and Dynamics

An orchard meadow combines fruit trees with a grassy, flower-rich understory, creating a unique habitat structure that promotes biodiversity. The vertical structure consists of two main layers: a canopy of fruit trees (like apple, pear, and cherry) and an understory of grasses, wildflowers, and small shrubs. The horizontal structure usually features irregular tree spacing to optimize sunlight for both tree growth and understory plants.

Orchard meadows are relatively low-disturbance ecosystems. The trees are often pruned, and the meadow floor may be managed by seasonal mowing or light grazing. These practices maintain open ground cover, prevent tree overcrowding, and support flowering plants that attract pollinators. This ecosystem encourages soil health and creates habitat for birds, insects, and small mammals, making it both productive and ecologically valuable.

Forest Development Objectives and management goals

1. Wood Production

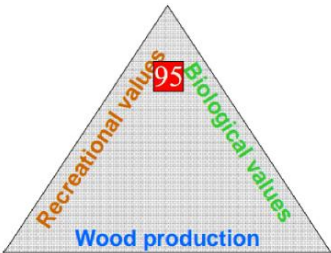
Orchard meadows offer economic value through sustainable fruit and wood production, as they generate income from diverse fruit yields. Additionally, they support ecotourism and recreational activities, attracting visitors interested in nature experiences, thus adding value through tourism and educational opportunities.

2. Biological Values

The orchard meadow supports pollinators by enhancing flowering plant diversity, attracts beneficial insects by retaining deadwood as habitat for small mammals and nesting birds, and promotes soil health through the use of organic mulches and minimal tilling to protect soil structure and invertebrate communities.

3: Recreational Values

The orchard meadow provides public access through carefully maintained trails that protect sensitive areas, enhances scenic beauty by pruning trees for optimal canopy height and open views, and serves as an educational site demonstrating sustainable, biodiversity-friendly agroforestry practices.



Occurrence

Typically found in well-drained, nutrient-rich soils with moderate sunlight and adequate moisture, is characterized by the coexistence of fruit-bearing trees and diverse herbaceous plants, fostering a unique ecosystem that supports both agricultural productivity and rich biodiversity.

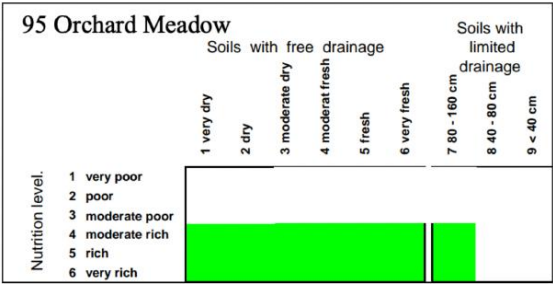


Figure 36: FDT 95 – Orchard Meadow